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Nitrate transport and transformation processes in unsaturated porous media

James A. Tindall^{a,*}, Robin L. Petrusak^b, Peter B. McMahon^c

^a*US Geological Survey, Water Resources Division, M.S. 413, Box 25046, Denver Federal Center, Denver, CO 80225-0046, USA*

^b*University of Colorado at Denver, Environmental Sciences Department, P.O. Box 173364, Denver, CO 80217-3364, USA*

^c*US Geological Survey, Water Resources Division, M.S. 415, Box 25046, Denver Federal Center, Denver, CO 80225-0046, USA*

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Abstract

A series of experiments was conducted on two contrasting agricultural soils to observe the influence of soil texture, preferential flow, and plants on nitrate transport and denitrification under unsaturated conditions. Calcium nitrate fertilizer was applied to the surface of four large undisturbed soil cores (30 cm diameter by 40 cm height). Two of the cores were a structured clay obtained from central Missouri and two were an unstructured fine sand obtained from central Florida. The cores were irrigated daily and maintained at a matric potential of -20 kPa, representative of soil tension in the rooting zone of irrigated agricultural fields. Volumetric water content (θ), concentration of nitrate-N in the soil solution, and nitrous oxide flux at the surface, 10, 20, and 30 cm were monitored daily. Leaching loss of surface-applied NO_3^- -N was significant in both the sand and the clay. In unplanted sand cores, almost all of the applied nitrate was leached below 30 cm within 10 days. Gaseous N loss owing to denitrification was no greater than 2% of the nitrate-N applied to the unplanted sand cores and, in general, was less than 1%. Although leaching was somewhat retarded in the clay cores, about 60% of the applied nitrate-N was leached from the unplanted clay soil in 5–6 weeks. Under unsaturated conditions, the clay had little to no tendency to denitrify despite the greater moisture content of the clay and retarded leaching of nitrate in the clay. The planted sand cores had surprisingly large gaseous N loss owing to denitrification, as much as 17% of the nitrate-N. Results from both the clay and sand experiments show that the dynamics of nitrate transport and transformation in unsaturated soils are affected by small, localized variations in the soil moisture

* Corresponding author.

content profile, the gaseous diffusion coefficient of the soil, the rate at which the nitrate pulse passes through the soil, the solubility of N_2O and N_2 and the diffusion of the gases through the soil solution, and development of a water content profile in the soil. Limited denitrification in the clay soil was due to a limited volume of soil available for infiltration after internal catchment and the development of denitrifying conditions resulting from the presence of an extensive macropore system.

1. Introduction

Nitrate loss from agricultural land through leaching and denitrification presents agricultural management and environmental problems. The use of manufactured nitrogen fertilizer has increased worldwide since the late 1940s. The application of nitrogen fertilizer in the United States increased from 500 000 metric tons (0.5 Tg) in 1945 to over 10 million metric tons (10 Tg) in 1976 (United States Department of Agriculture (USDA), 1977). Worldwide consumption in 1985 was 70.5 Tg (Food and Agriculture Organization of the United Nations (FAO), 1985) and Eichner (1990) predicted a consumption of about 94 Tg by 1997 based on World Bank estimates. During the past 20 years, the high cost of manufactured fertilizer has generated increased interest in land disposal of nitrogenous animal, municipal, and food processing wastes.

Water discharged from tile-drained fields and field run-off can leach nitrate into surface waters where it contributes to eutrophication in aquatic systems. Nitrate leached from agricultural lands also contaminates underlying aquifers. Numerous studies of water quality in agricultural areas of the United States have documented nitrate concentrations in ground water and surface water greatly in excess of the regulated United States drinking water standard of $10 \text{ mg l}^{-1} \text{ NO}_3^-$ (Keeney, 1982; Weil et al., 1990). Nitrate levels in household wells in some agricultural areas have reached acutely toxic concentrations (Keeney, 1982). Elevated nitrate concentrations in drinking water can cause methemoglobinemia in some livestock and in human infants under the age of 6 months. Nitrate consumption has also been implicated circumstantially in a greater incidence of stomach cancer in some localities (Lee and Dahab, 1992).

The transformation of N (N cycle) in soils is an integral part of nature but it can be affected greatly by agricultural and industrial operations. Usually, the alterations caused by agriculture and industry result from gains in soil N owing to fertilization and waste disposal operations. Forms of N added to the soil and N already present can transform readily from one by-product to another depending on the type and environmental conditions (a simple scheme of the N cycle and its simplified relationship to the unsaturated and saturated zones is given in Fig. 1). Transformation by-products of nitrogen can also be detrimental to the environment. Nitrous oxide emitted as an end product of denitrification contributes to the destruction of stratospheric ozone. Nitrous oxide is a stable molecule with no significant tropospheric sinks; it is transported slowly to the stratosphere where it either

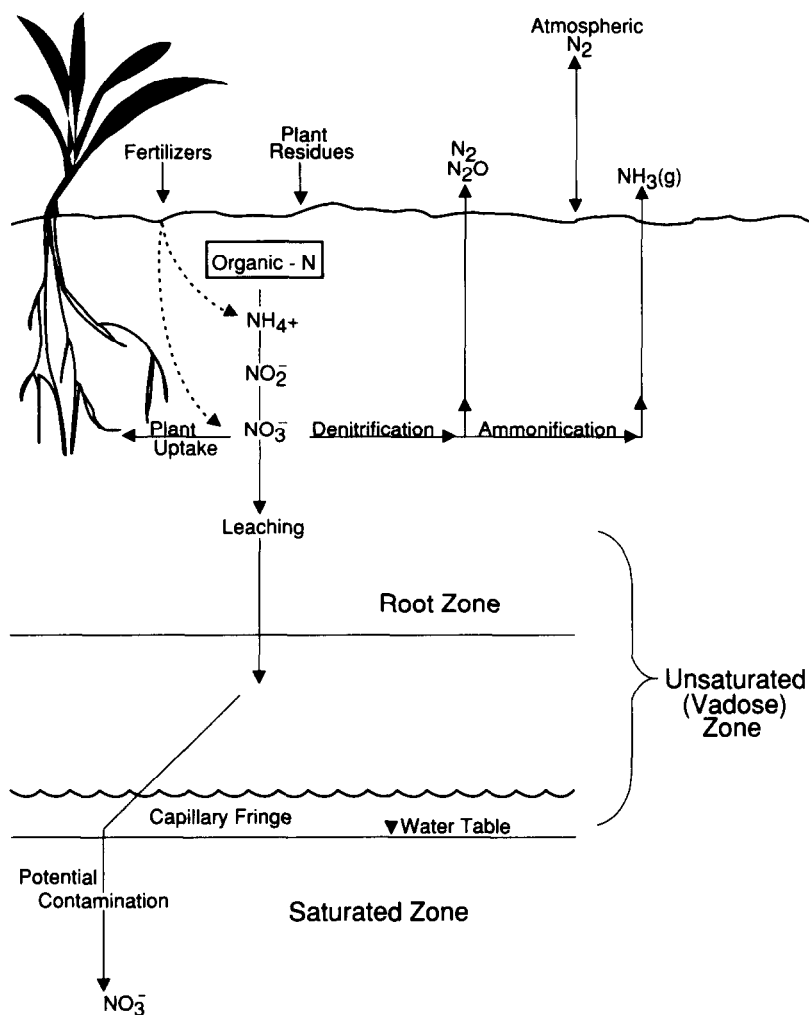
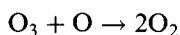
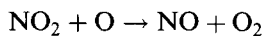
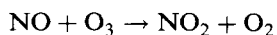


Fig. 1. Simplified scheme of the N cycle and basic relationship to the unsaturated and saturated zones.

photolyses to $\text{N}_2 + \text{O}_2$ or reacts with singlet oxygen to produce N_2 plus O_2 or NO . The NO thus produced reacts with stratospheric ozone (Finlayson-Pitts and Pitts, 1986)



The impact of biological N_2O emissions is expected to increase as the use of nitrogen fertilizers continues to increase worldwide. The magnitude of this impact is uncertain

because key variables are not known, accurately including the fraction of denitrification products emitted as N_2O , the residence time of N_2O in the troposphere, and the amount of denitrification occurring worldwide (Finlayson-Pitts and Pitts, 1986).

Requirements for denitrification in soil include a sufficiently large population of denitrifying bacteria, electron donor compounds, and the restricted availability of oxygen (Tiedje, 1988). The presence of organic compounds as electron donors and the restricted availability of oxygen are the most important controls on denitrification in soil systems. Oxygen availability in the unsaturated zone is controlled primarily by soil water content. Because the oxygen diffusion rate through water is about 104 times less than the diffusion rate through air, oxygen supply to microbially active soil microsites will decrease as volumetric water content increases and interconnected air-filled porosity decreases. Thus, denitrification activity, generally, is observed to increase as soil water content increases. Organic soil amendments which affect the C/N ratio, such as manure and crop residues increase denitrification activity (Comfort et al., 1990; Aulakh et al., 1991; Cabrera et al., 1994). The influence of living native plants or crops on denitrification in field soils is complex and previous studies show that the net effect depends upon factors including the availability of native soil carbon, soil pore size and water content, and the rate of nitrate supply to the root zone (Firestone, 1982). Despite 20 years of studies, the regulatory role of oxygen and organic carbon on denitrification in the unsaturated zone remains very difficult to quantify or model effectively.

This research was undertaken to obtain a better understanding of the fate of nitrate fertilizer in unsaturated porous media when applied to two contrasting agricultural soils. The specific objectives of this project were: (1) observe the leaching of nitrate and the evolution of nitrous oxide, a denitrification intermediate product, as a function of depth, time, and soil moisture content; (2) observe the influence of soil texture, preferential flow, and plants on nitrate transport and denitrification; (3) integrate the experimental data into an interpretation of the dynamics of nitrate transport and transformation under unsaturated conditions.

2. Materials and experimental methods

2.1. Soils

This research was conducted in the US Geological Survey Unsaturated Zone Soils Laboratory at the Denver Federal Center, Denver, Colorado on four large, intact soil cores (30 cm diameter by 40 cm height) obtained from two different sites and representing two soil extremes. Both soil types were from areas experiencing groundwater contamination from agriculture. An unstructured sand soil and a heavy clay soil characterized by cracks, burrows, and other macropores were selected as the two soil extremes. Both soils require large amounts of fertilizer to be economically productive. The two sand cores were obtained near Altamonte Springs, Seminole County, Florida from rapid, wastewater infiltration basin number 50 (wastewater treatment site) at the Reedy Creek Improvement District (Fig. 2(a)) and are

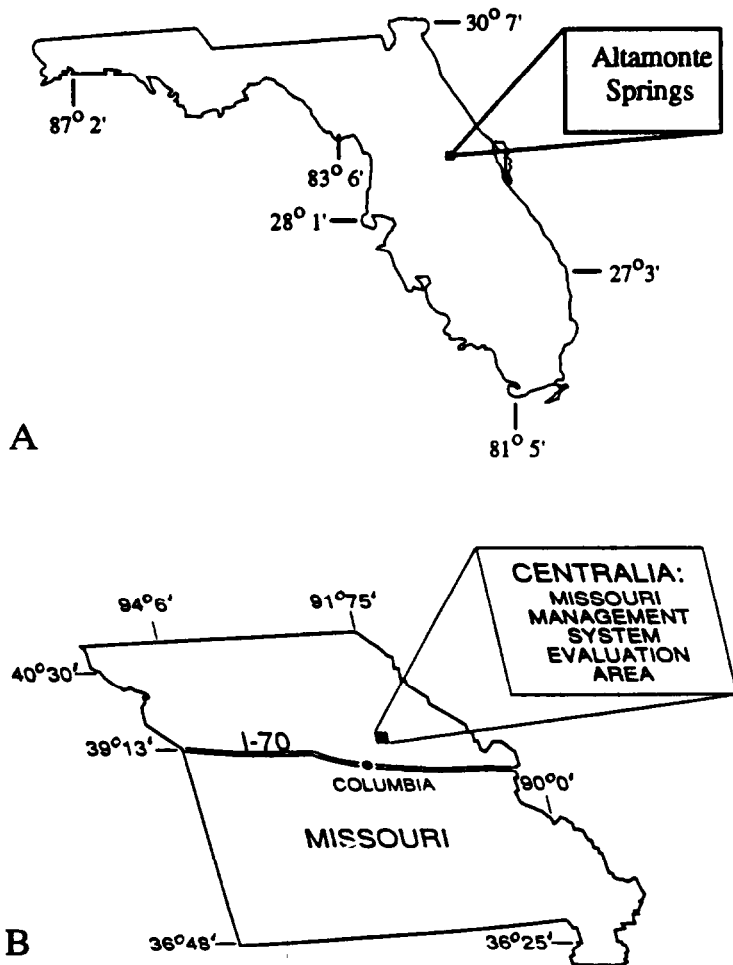


Fig. 2. Map indicating location of geographic areas used for core extraction.

representative of central Florida soils. The basins are not planted, but remain bare; however, the area immediately adjacent to the basins is planted in citrus orchards and has the same soil type. This soil is 93% sand and is classified as a Candler fine sand, hypothermic, uncoated, typic quartzsammments. The wastewater infiltration basin is conventionally tilled every other day, so the soil was moderately homogeneous and had no preferential flowpaths or macropores. The two clay cores were obtained from the Missouri Management System Evaluation Area near Centralia in Boone County, Missouri (Fig. 2(b)). This soil is classified as a fine montmorillonitic, mesic Udollic Ochraqualf Mollic albaqualf, silty loam. This soil is a well-structured, claypan subsoil common to the Midwest which contains preferential flowpaths owing to soil cracking and cleavage and the presence of root channels and burrows (Tindall and Vencill, 1993).

2.2. Experimental apparatus

Large undisturbed soil cores were used to approximate the physical and chemical properties of the field soil. Bouma (1979) and Bouma et al. (1979) showed that sample variability is reduced greatly by using cores with a minimum volume of 10 l. The volume of each soil core used in this study was 28.3 l. All cores were extracted from the 0 to 40 cm depth. The core extraction method and experimental apparatus is fully described by Tindall et al. (1992). A modification of the original apparatus is illustrated in Fig. 3. In brief, core extraction is accomplished by removing all loose litter from the site. The core extraction apparatus is pressed hydraulically into the soil. As the core barrel is pressed downwards in 5 cm increments, the soil around the exterior of the apparatus is removed to reduce soil resistance and soil compaction. The intact soil core is separated at the bottom interface of the apparatus with the in situ soil with a large knife. The intact soil core can then be removed from the apparatus, sealed, and transported back to the laboratory. Once back in the laboratory, the intact core is set atop a 1 bar, high flow porous ceramic plate (40 cm diameter), sealed, and instrumented. Gas sampling ports, tensiometers, soil solution samplers (suction lysimeters), and sets of TDR (time-domain reflectometry) rods for determination of volumetric water content were installed in each core at 10, 20, and 30 cm depths. The gas sampling ports were stainless steel tubes (38 cm long, 8 mm i.d., 9 mm o.d.) inserted horizontally into the cores. Each tube was perforated with four rows of 5 mm diameter holes about 1.2 cm apart as shown in Fig. 3. The tensiometers were automated with transducers attached to a data logger. The suction lysimeters were constructed of 1 bar high flow, tapered neck, round bottom porous ceramic cups attached to the end of schedule 80 pvc pipe (1.5 cm i.d.). Two tubes, one for sample discharge and one for air flow, were threaded through a rubber stopper at the top of the pvc pipe. The suction lysimeters were installed vertically into the cores as illustrated in Fig. 3 with the porous cup placed at the desired sample depth and sealed with bentonite near the soil surface to prevent piping. The sets of TDR rods consisted of two parallel stainless steel rods (21 cm long, 3.12 mm diameter) inserted horizontally. The TDR rods were connected at one end to a coaxial cable and were used to obtain a measurement of the apparent dielectric constant of the core material contacting and surrounding the rods. Volumetric water content of the soil was calculated from the TDR measurements.

Once the experimental apparatus was set up, each core was leached with 12 l of distilled water (≈ 1 pore volume) to displace existing water. A vacuum pump was attached to the cores and a suction of -20 kPa was applied. During the experiments, the soil cores were maintained at a matric potential of -20 kPa which is representative of the matric potential in the rooting zone of many irrigated vegetable crops (depth 0–30 cm). After completion of the experiments, a red dye (Keco-acid Crimson Red; Rite Industries, NC) was applied for staining of the cores. The red dye solution was ponded atop each core to a depth of 5 cm; head was maintained by a Marriott-apparatus. A magnetic stirrer was placed beneath the Marriott-apparatus to maintain equilibrium conditions in the red dye/tap-water mix. The dye was mixed at 2 parts solid per 20 parts water (by volume). The total amount of dye solution

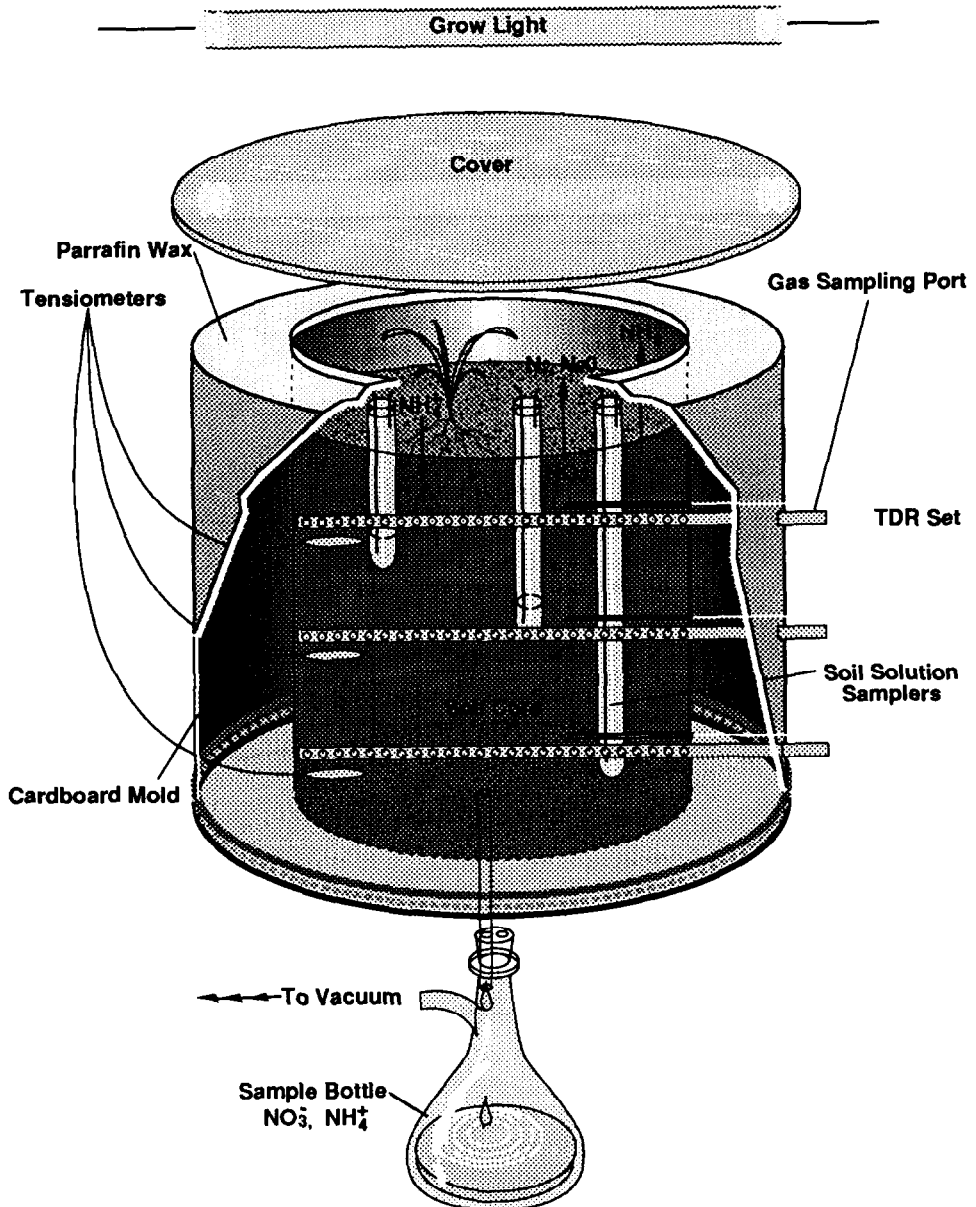


Fig. 3. Experimental apparatus.

applied to each core was equivalent to one pore volume (about 12 l). Each core was immediately sectioned at 5 cm depth increments to reduce transport owing to diffusion, and the total stained area versus soil core surface area was measured. The dyed areas were easily distinguishable from the surrounding soil.

2.3. Experimental method

The two sand cores were labelled cores 1 and 2 and the two clay cores were labelled core A and core B. At the beginning of each experiment, the cores were fertilized with 12.61 g calcium nitrate, $\text{Ca}(\text{NO}_3)_2 \cdot 4\text{H}_2\text{O}$, dissolved in 200 ml distilled water followed immediately with an application of 582 ml distilled water. This fertilization rate was equivalent to the recommended fertilization rate for tomatoes (235 kg ha⁻¹ applied in 2.5 cm precipitation) (Lorenz and Maynard, 1980). The dissolved fertilizer was sprayed evenly over the surface of the cores. After fertilization, the cores were watered daily with 400–600 ml distilled (and hence nitrogen-free) water sprayed evenly over the surface to simulate irrigation. However, during experiment 1, only about 180 ml of water were applied each day. It was realized later that this was not a sufficient rate and the rate was consequently adjusted upwards. Soil solution samples and soil gas samples were collected daily and volumetric water content was measured daily at 10 cm depth increments. The experiment was repeated six times on each sand core and only once on the clay cores. After each experiment, each core was flushed with one pore volume of distilled water (about 12 l) prior to beginning the next experiment to displace any remaining soil N before additions of N for the following experiment. Table 1 summarizes the experiments and shows total volumes of water applied. The evaporation rate was negligible (≈ 0.1 cm day⁻¹) and during the acetylene application period, evaporation was zero. The initial and upper boundary conditions were set at: initial — $\theta = f(z)$ at $z \leq 0$ and $t = 0$; upper boundary — $J = E_p$ for $z = 0$ and $t > 0$; where θ is the volumetric water content, z is depth (cm), t is time, J is flux (cm day⁻¹), and E_p is potential evaporation (cm day⁻¹).

Table 1
Experimental summary table

Experiment	Dates (1993)	Days	C ₂ H ₂	Plants	Water applied (ml/PV)
Sand #1	1/27–2/11	15	No	No	3346 ml (0.3 PV)
Sand #2	2/22–3/11	17	Yes	No	9046 ml (0.7 PV)
Sand #3	3/17–3/26	9	Yes	No	5282 ml (0.4 PV)
Sand #4	4/2–4/22	20	Yes	No	12332 ml (1.0 PV)
Sand #5	5/7–6/3	27	Yes	Tomato	15982 ml (1.3 PV)
Sand #6	6/5–6/20	15	Yes	Tomato	9182 ml (0.7 PV)
Clay core A	5/8–6/18	41	Yes	No	24700 ml (1.8 PV)
Clay core B	5/8–6/10	33	Yes	No	17500 ml (1.3 PV)

PV, pore volume.

Water collected in the outflow flask for each core was about equal to the water application rate and the volumes for the three samples extracted from the suction lysimeters from each depth ranged from about 10–30 ml per day for the sand cores and 21–30 ml for the clay cores. The sand cores were bare during experiment 1 through 4. Prior to experiment 5, the sand cores were planted with mature tomato plants. The clay cores were not planted.

Acetylene was applied because, at the concentration added, it inhibits nitrification; thus, in all but experiment 1, acetylene application was made by attaching a mylar chamber over the surface of each core when the fertilizer was applied and 12 l of acetylene (C_2H_2), (≈ 1 pore volume) was applied through the chamber. The acetylene-filled chamber remained over the cores until all of the acetylene penetrated the core (determined by a gas chromatograph). Including headspace and soil porosity, the amount of acetylene applied was as suggested by Davidson et al. (1986) and Cabrera et al. (1993). The chambers attached to the sand cores emptied in about 24 h. Acetylene was applied to the clay cores 72 h before fertilization to allow additional time for the C_2H_2 to penetrate the soil. Therefore, at least during the first few days, there should not have been any emission of N_2O by nitrifiers (due specifically to blockage of the N_2O reductase enzyme) (Klemmedtsson et al., 1988). Measurements of N_2O , therefore, provide an estimate of denitrification rates. Also, acetylene inhibition is most effective if incubation is limited to 7–10 days, diffusion of acetylene to microbially active soil microsites is optimized, and large volumes of acetylene are applied (Firestone, 1982; Davidson et al., 1986; Cabrera et al., 1994).

A 3 ml soil gas sample for N_2O analysis was taken daily from the surface of the cores and from the 10, 20, and 30 cm gas sampling ports. For each sampling, the sample ports were first purged by removing one pore volume of the sampling tube (20 ml) so that any old air in the tube would be replaced with soil air immediately outside the sampling port which should be in equilibrium with the surrounding soil for each sampling depth and a 3 ml gas sample was then withdrawn with a glass syringe. The gas sample was either analyzed immediately or was stored in an evacuated vial for analysis within 3 days. The soil gas samples were analyzed for N_2O with a gas chromatograph equipped with an electron capture detector. Carrier gas for the chromatograph was 95% argon and 5% methane. Nitrous oxide standards in air of 3.01, 30.1, and 300 ppm were used to calibrate the gas chromatograph measurements.

A water sample was collected daily from the outlet drainage flask and the soil solution samplers. Soil solution samples were obtained by applying -20 kPa tension to the suction lysimeters with a manual vacuum pump for 24 h; a solution sample was then discharged by applying positive pressure to the lysimeter with the same manual pump. The drainage flask was then drained, rinsed three times with distilled water, and reattached to the core apparatus. Approximately 15 ml of soil solution sample was frozen for future analysis for nitrate. Both the lysimeter and drainage flask samples are integrated samples for daily intervals. Soil solution samples and drainage flask samples were analyzed for NO_3^- -N using an Orion Model 95-12 ammonia electrode and a pH/mV meter.

Nitrous oxide flux was measured to determine the amount of applied N lost as the

gaseous end products of denitrification and was determined in the following manner. Diffusion is the primary mechanism for gas transport in soil. Gas flux in free air is expressed by Fick's Law of diffusion which must be coupled with the continuity equation to obtain Fick's Second Law

$$D^s \frac{\partial^2 C}{\partial z^2} = \frac{a \partial C}{\partial t} - q \quad (1)$$

where D^s is soil diffusivity ($L^2 T^{-1}$), a is the volumetric air content (unitless), C is concentration ($M L^{-3}$), and q is an algebraic source term. In this formulation, both solution and sorption are ignored. In this case, q represents the production rate of N_2O , as a concentration per unit time. At a given point, say the 10 cm depth, this production would diffuse toward the boundaries and would also go into storage in the vicinity of the node. Developing the continuity equation in finite-difference form to compute q at each node and each day, the mass per unit time of N_2O that has been added to storage in, say, the 10 cm probe, which is designated here as depth 1, would be expressed as

$$A \Delta z a \left(\frac{C_1^n - C_1^o}{\Delta t} \right) \quad (2)$$

where A is core area, Δz is incremental height (10 cm), and the superscripts n and o refer to the new and old time levels, respectively.

Nitrous oxide is fairly soluble and the gas phase can reasonably be assumed to equilibrate with the thin films of moisture in the unsaturated cores. Hence, the storage term needed to be modified to include the dissolved N_2O , using the equation

$$A \Delta z (a + \theta H) \left(\frac{C_1^n - C_1^o}{\Delta t} \right) \quad (3)$$

where H is the Henry's Law constant for N_2O in water in proper units ($H = 0.67$ at $20^\circ C$ and 83 kPa) Alberty (1987). The amount of q contributing to upward flux from that increment at time step n would be given by

$$A D_{0-1}^s \left(\frac{C_1^n - C_0^o}{\Delta z} \right) \quad (4)$$

where D is the diffusion coefficient operative between probe 1 and surface and C_0 is surface concentration, or 0. The flux at the previous time step is

$$A D_{0-1}^s \left(\frac{C_1^o - C_0^o}{\Delta z} \right) \quad (5)$$

and the average flux over the elapsed day is commonly taken as the average of the two,

$$A D_{0-1}^s \left[\frac{(C_1^n + C_1^o) - (C_0^n + C_0^o)}{2 \Delta z} \right] \quad (6)$$

Considering the downward flux from probe 1, based on concentrations in probes 1

and 2, and all terms (storage including solubility, flux up, and flux down) added together to equal q , in micrograms per unit time, the resulting equation (generalized by substituting i for node level) is

$$q = A\Delta z(a + \theta H) \left(\frac{C_i^n - C_i^o}{\Delta t} \right) + AD_i^s \left[\frac{(C_i^n + C_i^o) - (C_{i-1}^n + C_{i-1}^o)}{2\Delta z} \right] + AD_{i+1}^s \left[\frac{(C_i^n + C_i^o) - (C_{i+1}^n + C_{i+1}^o)}{2\Delta z} \right] \quad (7)$$

This equation is suitable for the first and second depths. For the third depth, the depth can be extended to the bottom of the column, making it 15 cm long, and D_4 set to zero (no diffusion downward across porous plate). However, the analysis is complicated by the fact that N_2O is being transported convectively by the infiltrating water as well as by diffusion.

Gas flux in the soil is calculated by modifying Eq. (1) by a gas tortuosity factor, ξ_g . The gas tortuosity factor accounts for the effect of the porous medium, the solid matrix, and the liquid-filled porosity on the gas diffusion pathways. D_{gs} is the soil

Table 2
Average volumetric water content

Experiment	Core #1 depth (cm)	Average volumetric water content (θ)	Core #2 depth (cm)	average volumetric water content (θ)
Sand #1	10	0.06	10	0.05
	20	0.11	20	0.06
	30	0.15	30	0.07
Sand #2	10	0.06	10	0.05
	20	0.14	20	0.09
	30	0.23	30	0.14
Sand #3	10	0.08	10	0.06
	20	0.20	20	0.12
	30	0.27	30	0.19
Sand #4	10	0.07	10	0.05
	20	0.17	20	0.09
	30	0.27	30	0.15
Sand #5	10	0.08	10	0.05
	20	0.20	20	0.10
	30	0.27	30	0.17
Sand #6	10	0.10	10	0.06
	20	0.27	20	0.14
	30	0.29	30	0.24
Clay #1	10	0.40	10	0.36
	20	0.38	20	0.33
	30	0.39	30	0.39

Table 3a
Nitrate-N leaching, sand cores (soil pressure = -20 kPa)

Depth ^a	Core 2									
	$\Delta\theta^b$	$\sum \Delta\theta^c$	PV ^d	PV at BT ^e	Hours	(NO ₃ -N) (peak) (ppm)	$\Delta\theta^b$	$\sum \Delta\theta^c$	PV ^d	(NO ₃ -N) (peak) (ppm)
<i>Sand experiment 1</i>										
10	0.4	0.4	0.03	0.04	48	1521	0.35	0.35	0.03	—
20	0.6	1.0	0.08	0.09	72	1053	0.4	0.8	0.07	—
30	0.9	1.9	0.16	0.19	216	1208	0.5	1.3	0.11	1881
40	1.2	3.1	0.25	0.27	264	477	0.5	1.8	0.15	661
<i>Sand experiment 2</i>										
10	0.4	0.4	0.03	—	—	—	0.35	—	—	—
20	0.7	1.1	0.09	0.12	48	3052	0.5	—	—	—
30	1.3	2.4	0.20	0.23	120	2133	0.8	1.7	0.14	4023
40	1.9	4.3	0.36	0.36	192	1444	1.2	2.9	0.24	2484
<i>Sand experiment 3</i>										
10	0.6	0.6	—	—	—	—	0.4	0.4	—	—
20	1.0	1.6	0.13	0.11	24	2207	0.6	1.0	0.08	2207

30	1.6	3.2	0.27	0.25	96	2236	1.0	2.0	0.17	0.25	96	1676
40	2.1	5.3	0.44	0.38	168	700	1.6	3.6	0.30	0.29	120	1552
<i>Sand experiment 4</i>												
10	0.5	0.5	0.04	–	–	–	0.4	0.4	0.03	–	–	–
20	0.9	1.4	0.12	0.11	24	1967	0.5	0.9	0.08	–	–	–
30	1.6	3.0	0.25	0.26	120	1200	0.8	1.7	0.14	–	–	–
40	2.0	5.0	0.42	0.45	216	701	1.0	2.7	0.23	0.26	120	747
<i>Sand experiment 5</i>												
10	0.6	0.6	0.05	–	–	–	0.4	0.4	0.03	–	–	–
20	1.0	1.6	0.13	0.25	96	646	0.5	0.9	0.08	0.25	96	413
30	1.7	3.3	0.28	0.35	144	2361	1.0	1.9	0.15	0.35	144	1215
40	2.1	5.4	0.45	0.40	168	1131	1.6	3.5	0.29	0.40	168	736
<i>Sand experiment 6</i>												
10	0.7	0.7	0.06	–	–	–	0.4	0.4	0.03	–	–	–
20	1.3	2.0	0.17	0.25	96	1102	0.7	1.1	0.09	0.16	48	1054
30	2.0	4.0	0.33	0.35	144	477	1.3	2.4	0.20	0.21	72	434
40	2.6	6.6	0.55	0.45	192	295	1.7	4.1	0.34	0.45	192	186

^a Depth in centimeters.^b Change in soil moisture content within core.^c Cumulative soil moisture.^d Pore volume equivalent within core.^e Pore volume at time of breakthrough (BT).

Table 3b

Nitrate-N leaching, clay cores (soil pressure = –20 kPa)

Core	Depth (cm)	Clay experiment					
		$\Delta\theta^a$	$\sum \Delta\theta^b$	PV ^c	PV at BT ^d	Hours	(NO ₃ -N) (peak) (ppm)
A	10	2.8	2.8	0.23	0.26	168	632
A	20	2.8	5.6	0.47	0.33	216	511
A	30	2.7	8.3	0.69	0.07	24	568
A	40	2.8	11.1	0.93	0.26	168	228
B	10	2.5	2.5	0.21	0.07	24	1057
B	20	2.4	4.9	0.41	0.09	48	7.3
B	30	2.5	7.4	0.62	0.21	144	421
B	40	2.5	9.9	0.83	0.30	192	768

^a Change in soil moisture content within core.^b Cumulative soil moisture.^c Pore volume equivalent within core.^d Pore volume at time of breakthrough.

gas diffusion coefficient (Jury, 1991)

$$D_g^s = \xi_g D_g^a \quad (8)$$

The binary gas diffusion coefficient for N₂O in air was calculated using the following equation from Bird et al. (1960)

$$\frac{PD_{ab}}{(P_{ca}P_{cb})^{1/3}(T_{ca}T_{cb})^{5/12}\left(\frac{1}{M_a} + \frac{1}{M_b}\right)^{1/2}} = a\left(\frac{T}{\sqrt{T_{ca}T_{cb}}}\right)^b \quad (9)$$

where D_{ab} is a binary gas diffusion coefficient, P is atmospheric pressure, P_c is the critical pressure, T_c is the critical temperature (K), $M_{a,b}$ is the molecular weight of gas a , gas b , T is the experimental temperature (K), $a = 2.745 \times 10^{-4}$ (for non-polar gas pairs), $b = 1.823$ (for non-polar gas pairs).

By way of strict definition, the critical state is the state of temperature (T_c) and pressure (P_c) at which the gas and liquid phases become so nearly alike that they can no longer exist as separate phases. The diffusion coefficient calculated for N₂O in air at the experimental temperature (20°C) and pressure (–20 kPa) was $D_{N_2O}^a = 0.182 \text{ cm}^2 \text{ s}^{-1}$. This compares well when using chlorofluorocarbons as a reference check. The soil gas diffusion coefficient was calculated by applying the gas tortuosity model, ξ_g , derived by Millington (1959)

$$\xi_g = a^{10/3} \phi_T^{-2} \quad (10)$$

where a is volumetric air content and ϕ_T is the total porosity.

Air content was determined by subtracting measured volumetric water content from total porosity. For the sand cores, the calculated soil gas diffusion coefficient,

D_g^s ranged from $0.04988 \text{ cm}^2 \text{ s}^{-1}$ for $a = 0.42$ to $0.00069 \text{ cm}^2 \text{ s}^{-1}$ for $a = 0.12$. For the clay cores, the soil gas diffusion coefficient ranged from $0.00215 \text{ cm}^2 \text{ s}^{-1}$ for $a = 0.17$ to $0.00004 \text{ cm}^2 \text{ s}^{-1}$ for $a = 0.05$. Nitrous oxide flux across the surface of the cores and at the 10, 20, and 30 cm, depths was calculated using a FORTRAN program to solve Eq. (7). The N_2O concentration measured in ppm was converted to micrograms per cubic centimeter by the following conversion factor (Finlayson-Pitts and Pitts, 1986): at 0.8236 atm, $1 \text{ ppm} = (3.36 \times 10^{-5}) \times (\text{molecular weight})$ thus, concentration ($\mu\text{g cm}^{-3}$) is $(\text{ppm})(3.35 \times 10^5)(44.013)$. For example, 48 h after fertilizer application, measured N_2O concentration was 1.1 ppm ($1.63 \times 10^{-4} \mu\text{g cm}^{-3}$) in the 10 cm probe. Consequently, $\theta = 0.36$, $\phi_T = 0.48$, $a = \phi_T - \theta = 0.48 - 0.36 = 0.12$, $\xi_g = (0.12)^{3.333} / (0.48)^2 = 0.0037$, $D_{\text{N}_2\text{O}}^s = (0.0037) \times (0.182 \text{ cm}^2 \text{ s}^{-1}) = 6.734 \times 10^{-4} \text{ cm}^2 \text{ s}^{-1}$, $J_{\text{N}_2\text{O}} = \text{Eq. (7)} (\Delta z = 10 \text{ cm})$, $J_{\text{N}_2\text{O}} = 0.30 \mu\text{g N}_2\text{O h}^{-1}$.

3. Results and discussion

3.1. Nitrate leaching: sand cores

The 1496 mg N applied as nitrate at each fertilizer application simulated the standard N application rate for tomatoes. The initial concentration of $\text{NO}_3^- \text{N}$ dissolved in 200 ml H_2O was 7480 mg N l^{-1} . The concentration of $\text{NO}_3^- \text{N}$ dissolved in the total volume of water applied initially was 1913 mg N l^{-1} (measured). Depending upon the extent to which the 200 ml of dissolved fertilizer mixed with the 582 ml of water applied immediately afterward, the concentration of $\text{NO}_3^- \text{N}$ applied at time 0 h ranged between 1913 and 7480 mg l^{-1} . The unsaturated hydraulic conductivity, $K(\theta)$ (determined by a one-dimensional form of the Richard's equation using θ values obtained by the TDR — these values correspond well with those obtained utilizing Millington and Quirk (1961)), for core 1 was 0.65 cm day^{-1} ; $K(\theta)$ for core 2 was 0.74 cm day^{-1} . Table 2 shows that sand core 2 had a lower volumetric water content (at all depths for all experiments) than core 1, consistent with the greater hydraulic conductivity of core 2. Very few soil solution samples were obtained from the suction lysimeters placed at 10 cm in either sand core. When the 200 ml of fertilizer solution was applied, followed by 600 ml of water as a chaser, it is likely that rapid redistribution took place. This, coupled with assumed piston displacement of the residual soil moisture (or any underlying moisture), assumes that the chaser will push the concentrated solution to a depth below the sampler in a very short time (possibly on the order of several minutes), depending on the duration of water application. Thus, the water moved through the cores too quickly to be captured in the 10 cm lysimeter. This problem seems to have been exacerbated in core 2 which also had a greater hydraulic conductivity. Consequently, few solution samples were obtained from the 20 cm depth in experiments 1, 2, and 4 from this core.

Tables 3(a) and (b) summarize the $\text{NO}_3^- \text{N}$ leaching results. Conceptually, if no sorption or consumption, are assumed and piston displacement is operative, then the peak concentration should occur when the centroid of the 200 ml fertilizer slug is pushed past a given probe by adding a volume of water equal to that stored between

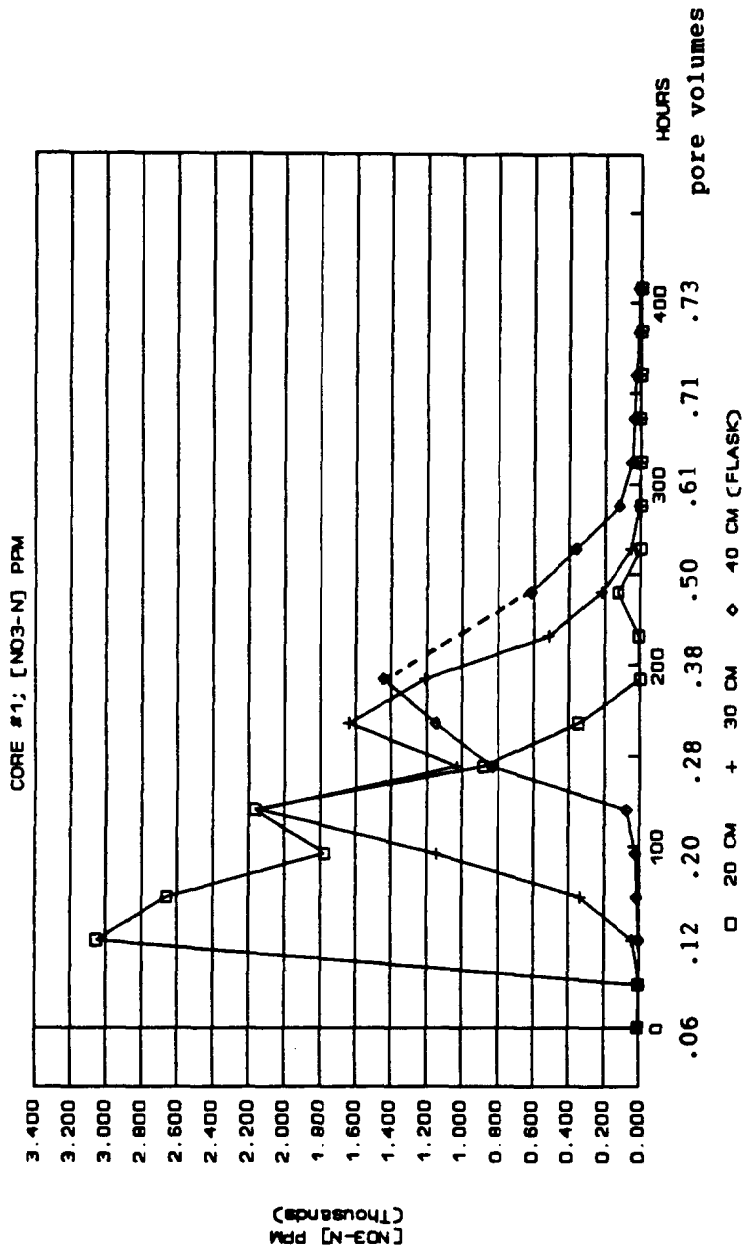


Fig. 4. NO₃⁻-N vs. time and pore volumes of water applied; sand experiment no. 2.

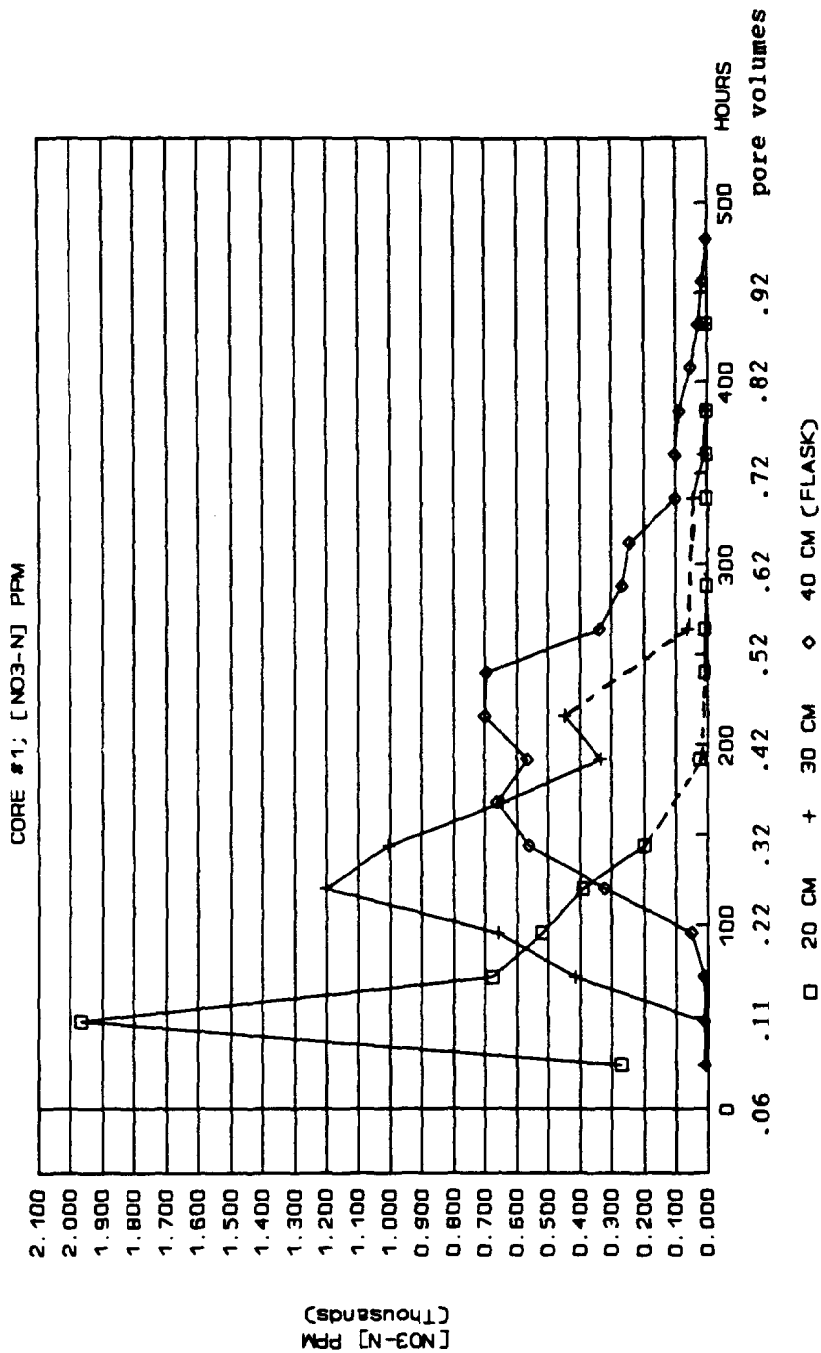


Fig. 5. $\text{NO}_3\text{-N}$ vs. time and pore volumes of water applied; sand experiment no. 4.

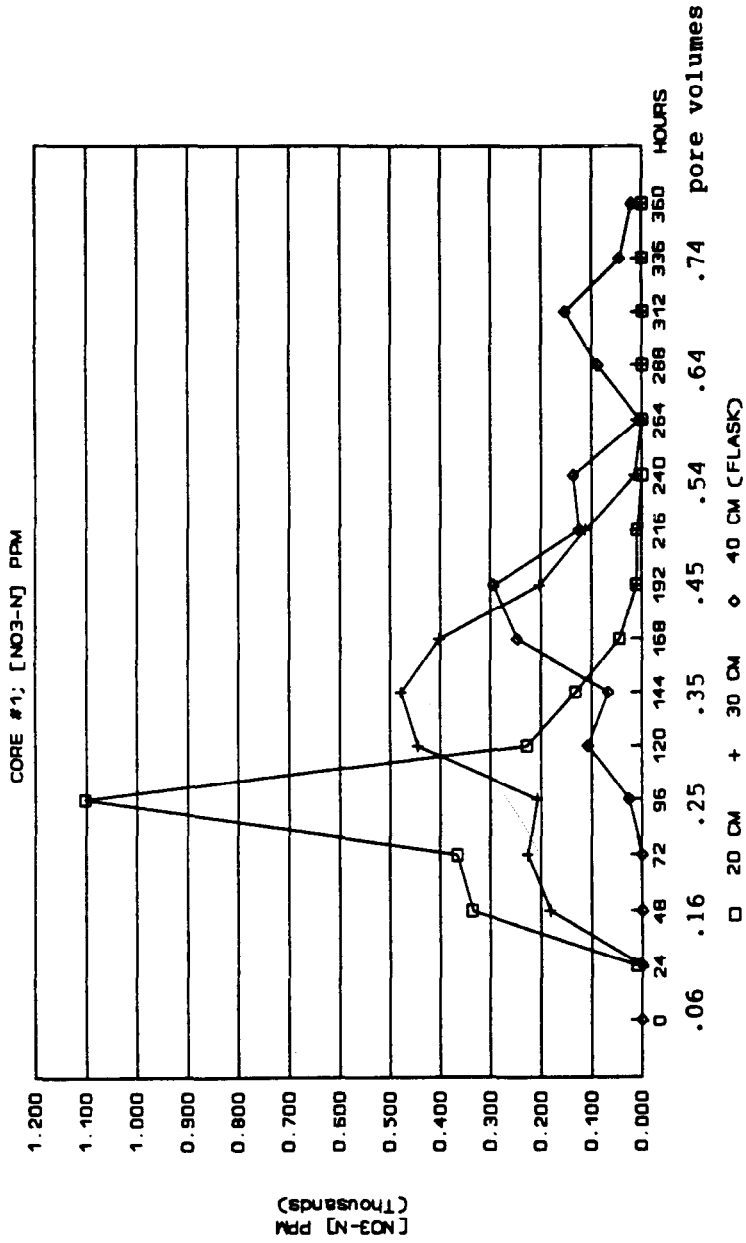


Fig. 6. $\text{NO}_3\text{-N}$ vs. time and pore volumes of water applied; sand experiment no. 6.

the surface and that depth. Consequently, the volume of water added from the start of the experiment until the peak occurred at a given probe was normalized by dividing by $\sum \theta \Delta z$, where the summation was over the probe increment number, starting at the top of the column. Utilizing this method, there was very close (sometimes exact) agreement between water flushed at peak concentration and the average residual water content during the experiment; this supports the conceptual model of no or very little sorption and of piston displacement through the sand. However, the tracer peak broke through before one pore volume of water had been displaced for all depths (Figs. 4–6), which is in keeping with other column studies of unsaturated transport in which only a part of the initial water is displaced, with dead-end pore water exchanging slowly with the mobile water. In the unplanted sand cores, the NO_3^- -N pulse passed 20 cm at about 0.11 pore volume (24–72 h); the NO_3^- -N peak passed 30 cm after about 0.23 pore volumes and passed 40 cm after 0.27 to 0.45 pore volumes. Figs. 4 and 5 are representative of NO_3^- -N concentration versus time for the unplanted sand experiments. At 20 cm, the peak concentration of NO_3^- -N was within the range of the concentration of the initial fertilizer application; this suggests little nitrogen loss owing to denitrification. At 30 cm and 40 cm, the concentration of NO_3^- -N was successively lower owing to dispersion, dilution and possible N loss to denitrification.

In the planted sand cores (experiments 5 and 6), the peak concentration of NO_3^- -N reached 20 cm after 0.25 pore volumes, 30 cm after 0.35 pore volumes, and 40 cm after 0.40–0.45 pore volumes. Nitrate movement through the rooting zone (0–30 cm) in the planted cores required slightly more water than in the unplanted cores because of plant water use. Fig. 6, a representative NO_3^- -N breakthrough plot for the planted cores, shows significantly lower peak concentrations (comparing Figs. 4 and 5) at 20, 30, and 40 cm owing to additional nitrate uptake by the plants and possibly to denitrification losses.

3.2. Nitrate leaching: clay cores

The unsaturated hydraulic conductivity, $K(\theta)$ for core A was 0.36 cm day^{-1} ; $K(\theta)$ for core B was 0.53 cm day^{-1} . Figs. 7 (core A) and 8 (core B) illustrate the effect of preferential flow in the clay soil. Peak NO_3^- -N breakthrough in core A (Fig. 7) occurred almost immediately at 30 cm (24 h, 0.07 pore volume (PV)) while it took 48 h before a solution sample could be obtained at 10 and 20 cm. The peak concentration of NO_3^- -N at 10 and 40 cm occurred at 168 h or 0.26 PV. It appears that water was channelled to 30 cm via preferential flow and, as suggested by the TDR data, became ponded in a tight zone in core A somewhat below the 30 cm depth. Staining of the cores after the experiment indicated no edge flow (see Figs. 9 and 10). After 7 days, the solution ponded at 30 cm broke through to 40 cm and the NO_3^- -N peak was evident. A similar time was required for surface-applied NO_3^- -N to breakthrough at 10 cm via less permeable flow paths. For core B, Fig. 8 shows rapid nitrate breakthrough at 10 and 20 cm, after 0.07 PV and 0.09 PV, respectively. This breakthrough compares well with the stained area at these depths for this core (Fig. 10). Nitrate breakthrough occurred at 30 cm after 0.21 PV and at 40 cm after 0.30 PV. In terms of

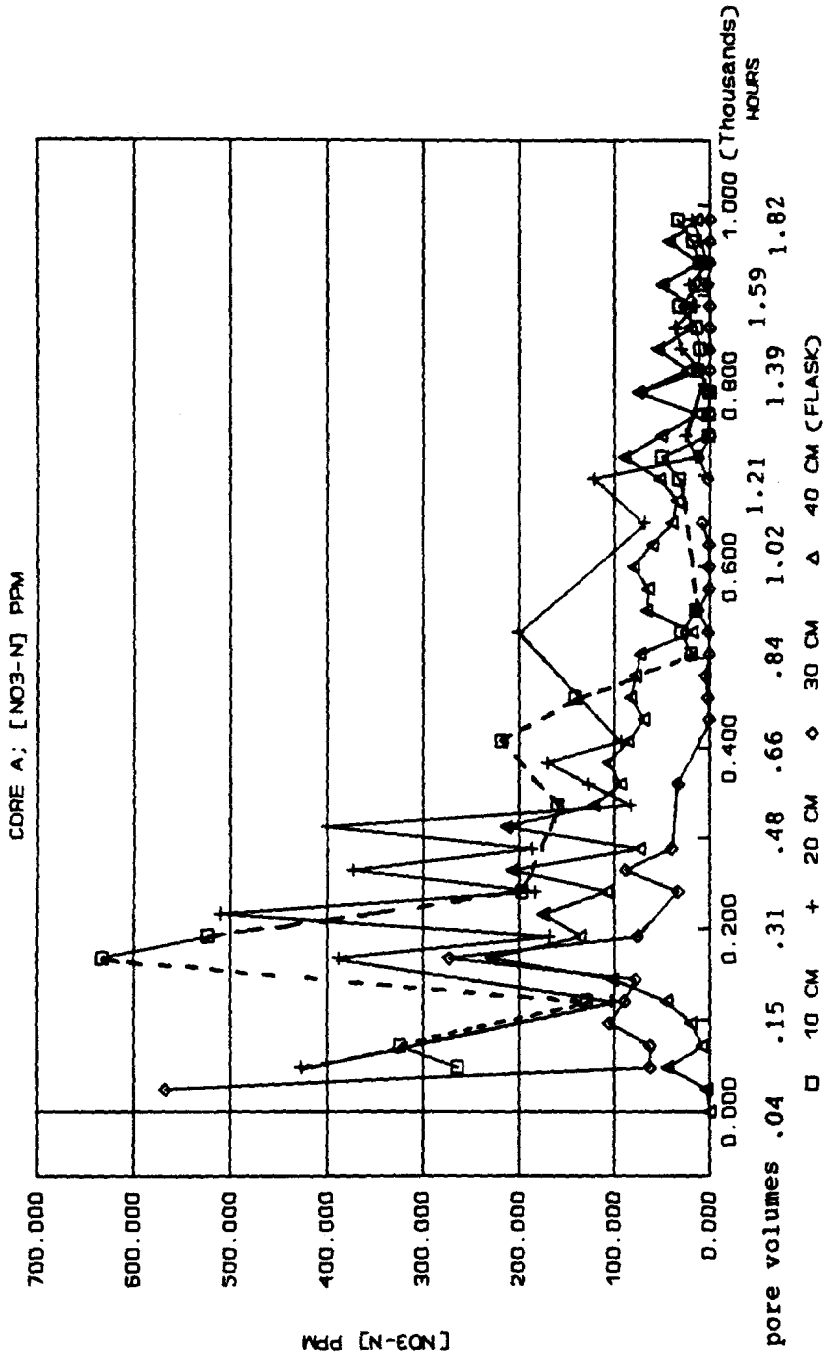


Fig. 7. NO₃-N vs. time and pore volumes of water applied; core A.

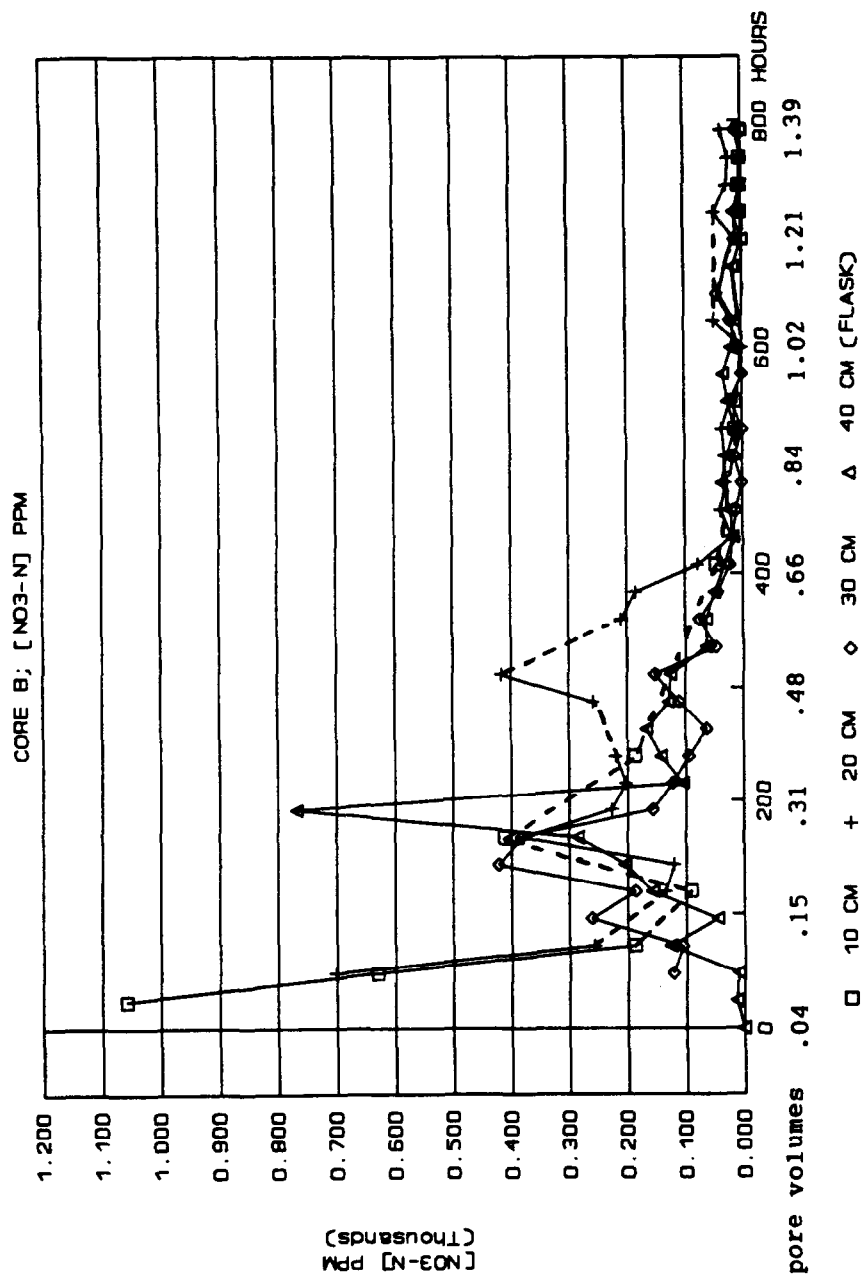


Fig. 8. NO₃⁻-N vs. time and pore volumes of water applied; core B.

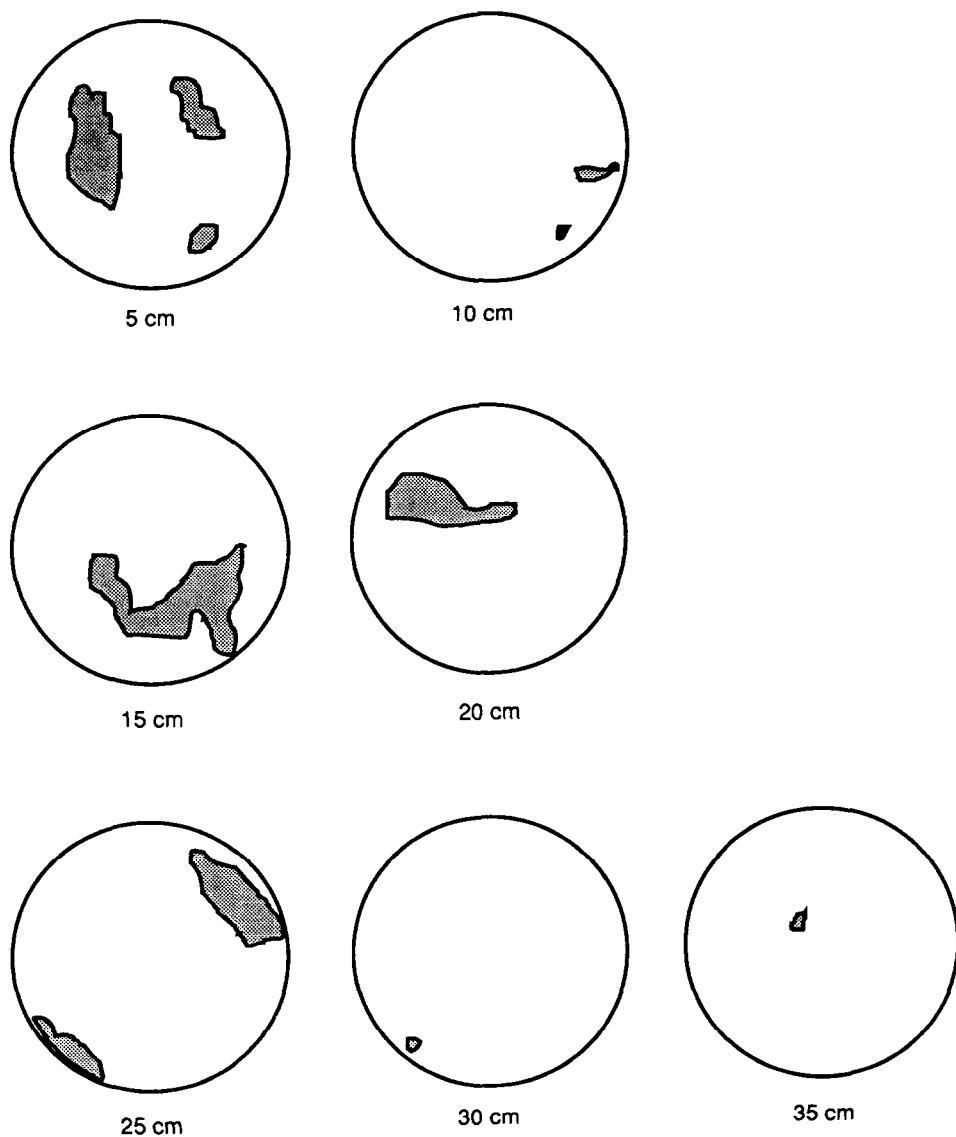


Fig. 9. Stained areas for core A at 5 cm increments. Shaded areas represent preferential flowpaths (areas of lower bulk density than surrounding soil matrix); dark areas represent macropores, i.e. worm holes, faunal tunnels, etc. (diagrams are 1/6th scale).

pore volumes applied, the NO_3^- -N pulse passed through core B more quickly than through the sand cores. The time required for NO_3^- -N to breakthrough in core B was comparable with the sand cores. The leaching results suggest that core B had a more permeable network of preferential flow paths than core A, and that core B had no clearly visible tight zone at the 30 cm to 40 cm depth (Fig. 10). The jagged character of

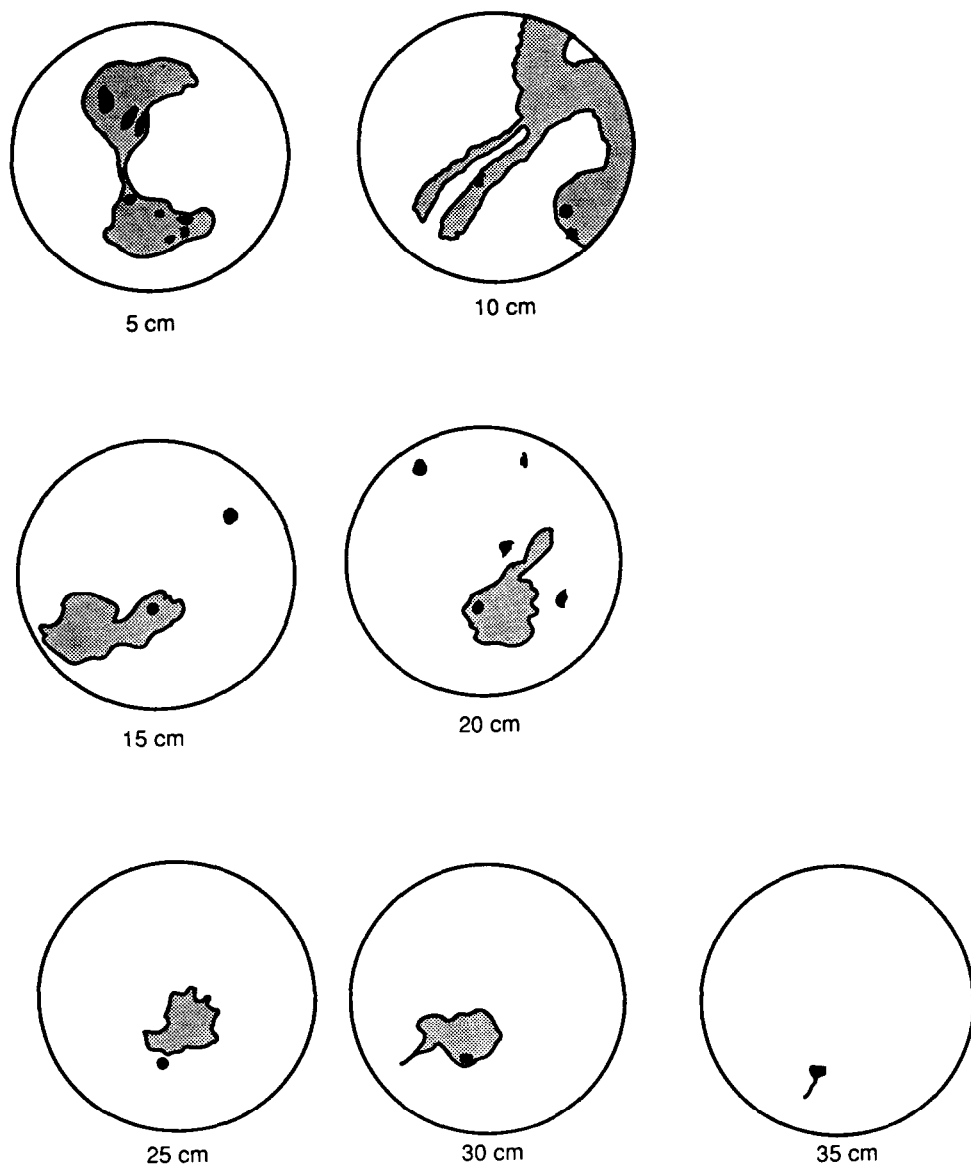


Fig. 10. Stained areas for core B at 5 cm increments. Shaded areas represent preferential flowpaths (areas of lower bulk density than surrounding soil matrix); dark areas represent macropores, i.e. worm holes, faunal tunnels, etc. (diagrams are 1/6th scale).

the nitrate breakthrough curves at 10, 20, and 40 cm for core A (Fig. 9) suggests preferential flow via fewer flow paths or poorly interconnected flow paths. This can be seen in Fig. 9 where there is a larger stained area at 5 cm than at 10 cm, a larger stained area at 15 cm than at 10 cm, followed by a reduced area at the remaining depths.

Thus, there is a non-even distribution from one depth to the next. Here, preferential flowpaths are defined as areas of a lower bulk density than the surrounding soil (represented as shaded areas in Figs. 9 and 10) while true macropores are defined as visually observable holes such as root or worm holes and faunal tunnels (represented as dark or black areas in Figs. 9 and 10). In this study, macropores were much less numerous and voluminous than preferential flowpaths.

3.3. Nitrate leaching: N recovery

Daily NO_3^- -N recovery was determined by multiplying the measured NO_3^- -N concentration of the drainage water by the daily drainage volume. The results are summarized in Table 4 and Figs. 11, 12, and 13. Note that NO_3^- -N recovery for sand experiments 2, 4, and 5 is greater than the 1496 mg applied in each fertilization. Also, since the cores were leached after each experiment, there is no accumulation of high nitrate concentrations within the flask at the start of each new experiment (Figs. 12 and 13). A recovery greater than 100% (Table 4) may be due to N coming from mineralization of soil organic matter. However, to check for this, one would need to have a control with no additions of nitrate; this was not done in this experiment. Also, it is possible that part of the excess nitrate recovered in the experiment could be due to nitrate applied, despite leaching after each experiment; sometimes, nitrogen is

Table 4
Nitrate leaching summary: total N recovery

Experiment	Core	At time of peak N_2O flux			At end of experiment		
		Time (h)	Cum.N leached (mg)	N recovered (%)	Time (h)	Total N leached (mg)	N recovered (%)
Sand #1	1	No peak N_2O flux			360	605	40
	2				360	595	40
Sand #2	1	168	1147	77	408	2786	101
	2	168	2249	101	408	2484	103
Sand #3	1	216	1099	73	216	1099	73
	2	216	1491	100	216	1491	100
Sand #4	1	240	1721	102	480	2164	102
	2	240	2129	103	480	2293	103
Sand #5	1	216	1907	103	648	2356	102
	2	216	1443	96	648	1673	104
Sand #6	1	240	379	25	360	458	31
	2	240	558	37	360	705	47
Clay	A	240	268	18	984	877	59
	B	240	618	41	792	963	64

1496 mg N applied as NO_3 at each experiment.

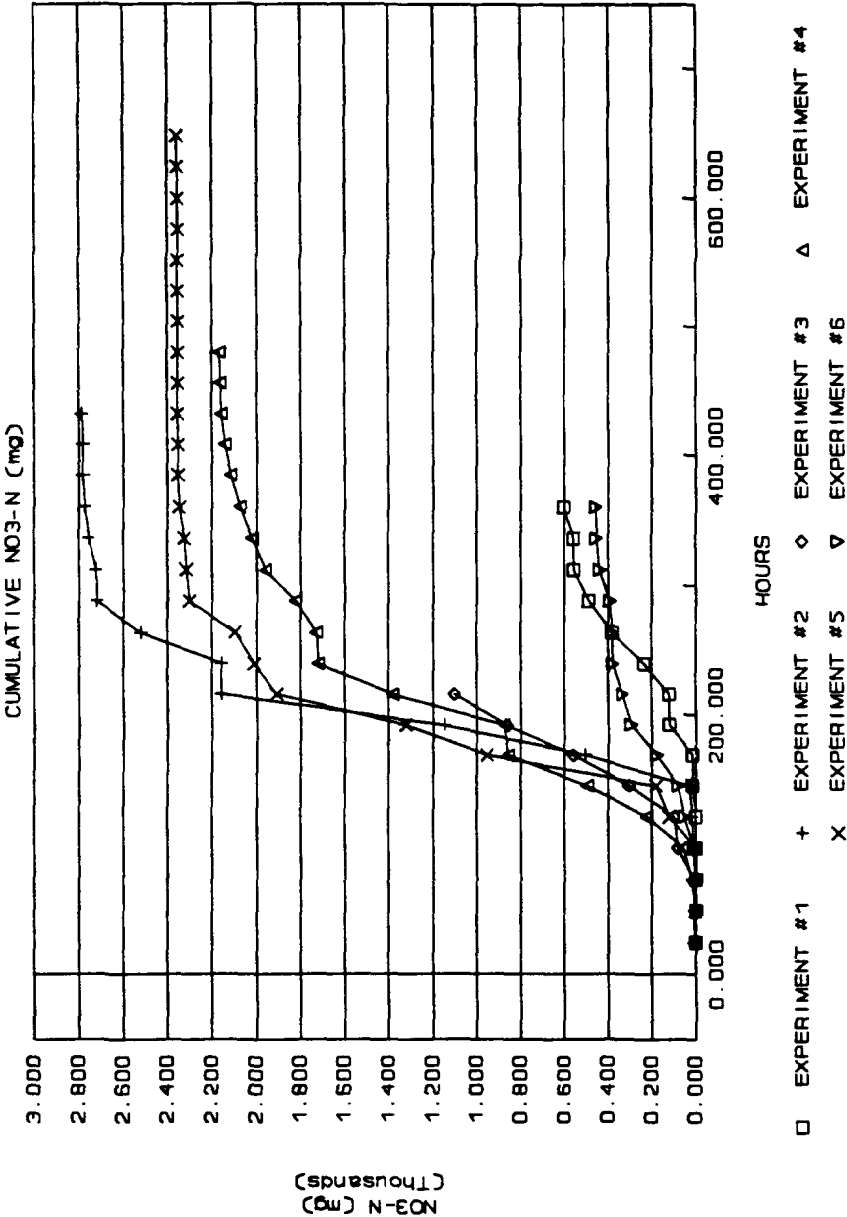


Fig. 11. Cumulative NO₃-N recovered (mg N); core 1, experiments 1–6 (40 cm (flask)).

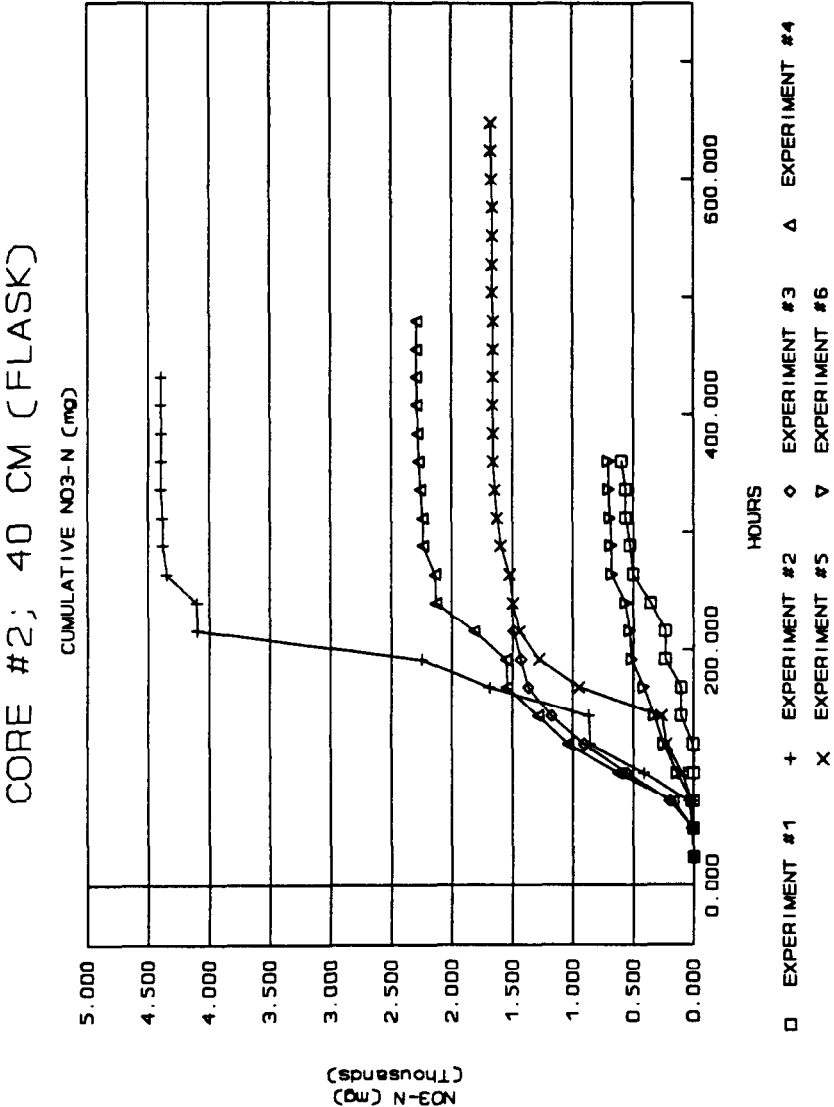


Fig. 12. Cumulative NO₃-N recovered (mg N); core 2, experiments 1-6 (40 cm (flask)).

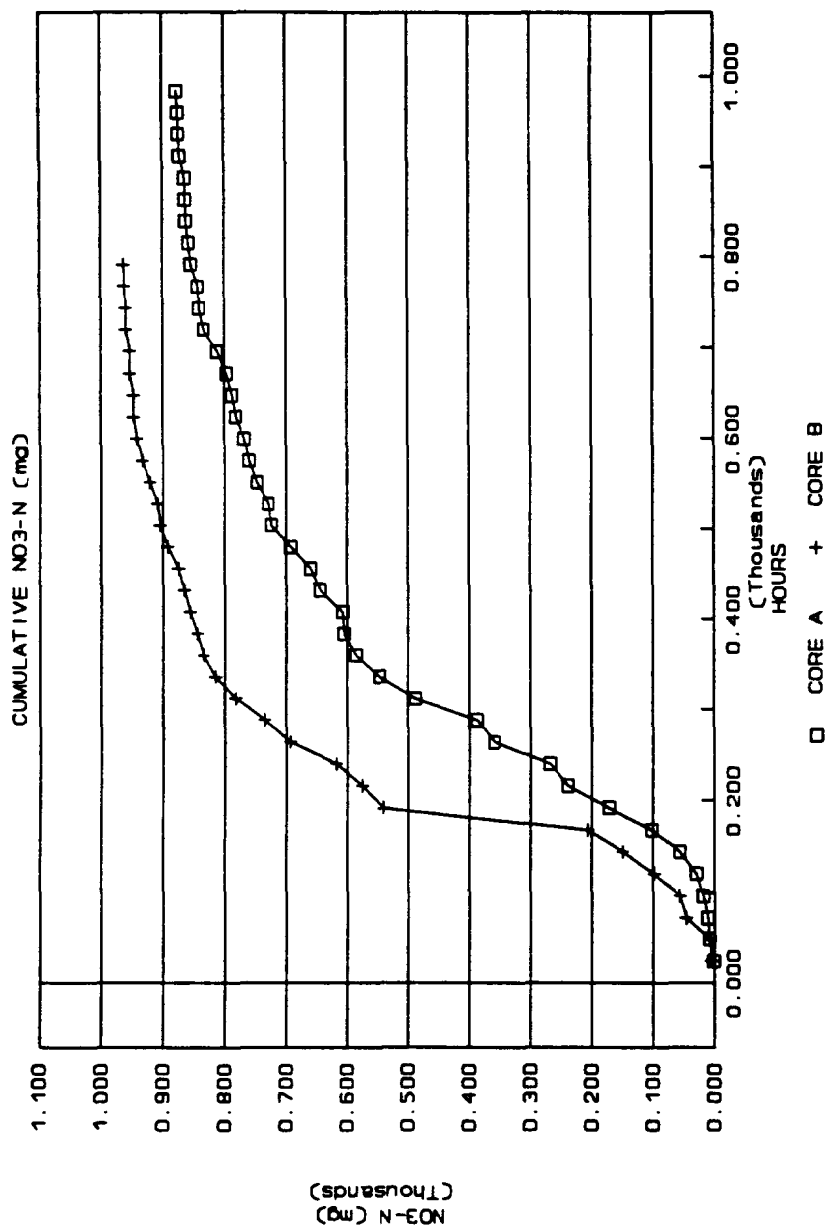


Fig. 13. Cumulative NO₃⁻-N recovered (mg N); cores A and B (40 cm (flask)).

immobilized and then remineralized at a later time. Despite this, it appears that none of the applied nitrate was retained by the sandy soil except in experiments 1 and 6.

Based on relatively good agreement between pore volumes of stored water versus pore water throughflow for sand column 1, experiment 1, there appears to be very little retardation of nitrate through the column, but the peak is substantially attenuated. Any sorption process that results in peak attenuation should also result in retardation. Hence, it would appear that the apparent nitrate loss was due to transformation. Experiment 1 differs in slower rates of irrigation, lack of acetylene application, and the fact that the column had no previous large quantities of nitrate flushed through it when compared with subsequent experiments. It is likely that the core originally contained sufficient organic carbon to transform about a gram of nitrate. Under these conditions there would have been too little organic carbon in the core to transform the slugs of nitrate in subsequent experiments. Consequently, experiment 1 behaved quite differently from experiments 2–5.

In sand experiment 6, the tomatoes planted at the outset of experiment 5 were well established and plant uptake of nitrate was evident in both sand cores. Also, as discussed in the next section, denitrification activity during experiment 6 resulted in gaseous N loss of 10% to over 17% in core 1. Table 4 shows that approximately 40% of the NO_3^- -N applied to the clay cores was not recovered. Since the clay cores were not planted, this deficit can not be attributed to plant uptake. Gaseous N loss owing to denitrification was insignificant in the clay cores hence, it is likely the remaining NO_3^- -N was retained or immobilized within the cores.

Leaching of nitrate is often the most important mechanism for nitrogen removal from irrigated field soils other than N removal by plant uptake (Legg and Meisinger, 1982). Rapid breakthrough and greater leaching of the nitrate applied to the clay cores in this study was expected due to the cracks, root channels, and other macropores visible on the surface of the cores. In fact, more leaching occurred in the sand cores than in the clay cores, despite the prevalence of stained areas in the clay cores which would normally indicate a greater capability for leaching (Figs. 9 and 10). In the unplanted sand cores, from 73% to over 100% of applied N was leached to the 40 cm depth within 10 days of application except in experiment 1. In the sand cores planted with mature tomato plants, the plants intercepted a significant portion of the applied N; nevertheless, 25–37% of N was lost below the rooting zone within 10 days except for experiment 5. In this experiment, it is likely that the plant rooting system was not well developed and, thus, the leaching behavior was similar to experiments 2, 3, and 4 (see Table 4). Less nitrogen was leached from the clay cores than the sand cores despite the apparent preferential flow along macropores in the clay soil. Clay core B lost 41% of applied N within 10 days (Fig. 13). Clay core A, with significantly less preferential flow than core B, lost only 18% of applied N in 10 days (Fig. 13). Sixty four percent of the nitrogen applied to core B was removed by leaching after 33 days; 59% of the nitrogen was removed from core A by leaching after 41 days (Table 4). Presumably, less nitrate would have leached from the clay cores had they been planted.

Table 5
Peak N₂O flux during sand experiments under acetylene inhibition

Core #	Depth (cm)	Sand experiment #2			Sand experiment #3			Sand experiment #4			Sand experiment #5 (plant)			Sand experiment #6 (plant)		
		Peak N ₂ O flux ($\mu\text{g h}^{-1}$)	Peak flux time (h)	(days)	Peak N ₂ O flux ($\mu\text{g h}^{-1}$)	Peak flux time (h)	(days)	Peak N ₂ O flux ($\mu\text{g h}^{-1}$)	Peak flux time (h)	(days)	Peak N ₂ O flux ($\mu\text{g h}^{-1}$)	Peak flux time (h)	(days)	Peak N ₂ O flux ($\mu\text{g h}^{-1}$)	Peak flux time (h)	(days)
1	0	149	144	6	150	144	6	204	72	3	248	192	8	908	144	6
1	10	265	144	6	343	192	8	214	216	9	423	192	8	862	192	8
1	20	115	144	6	201	216	9	95	168	7	339	192	8	724	192	8
1	30	37	120	5	495	216	9	693	216	9	727	144	6	2914	192	8
2	0	58	96	4	131	216	9	100	168	7	126	120	5			
2	10	64	120	5	272	192	8	196	192	8	182	192	8			
2	20	86	144	6	393	192	8	172	216	9	310	216	9	483	192	8
2	30	72	120	5	335	192	8	112	192	8	279	144	6	589	216	9

Table 6
Cumulative N₂O flux during sand experiments under acetylene inhibition

Core #	Depth (cm)	Sand experiment #2			Sand experiment #3			Sand experiment #4			Sand experiment #5			Sand experiment #6		
		Cum. N ₂ O flux (mg)	Peak flux time (h)	(days)	Cum. N ₂ O flux (mg)	Peak flux time (h)	(days)	Cum. N ₂ O flux (mg)	Peak flux time (h)	(days)	Cum. N ₂ O flux (mg)	Peak flux time (h)	(days)	Cum. N ₂ O flux (mg)	Peak flux time (h)	(days)
1	0	10.21	144	6	13.01	144	6	9.59	72	3	19.28	192	8	40.65	144	6
1	10	19.57	144	6	34.41	192	8	5.78	216	9	40.53	192	8	64.88	192	8
1	20	9.73	144	6	12.79	216	9	8.74	168	7	23.20	192	8	40.83	192	8
1	30	2.11	120	5	17.62	216	9	30.26	216	9	29.28	144	6	165.95	192	8
2	0	1.38	96	4	14.76	216	9	5.18	168	7	6.35	120	5	42.95	240	10
2	10	6.80	120	5	14.03	192	9	11.30	192	8	15.01	192	8	16.81	192	8
2	20	7.00	144	6	20.15	192	8	15.62	216	9	23.75	216	9	24.02	192	8
2	30	3.35	120	5	14.56	192	8	9.12	192	8	6.69	144	6	14.13	216	9

3.4. N_2O flux: sand cores

Sand experiment 1 had significantly lower N_2O flux than experiments 2 through 6 because no acetylene was applied to the cores and the water content of both cores during experiment 1 was much lower than the subsequent sand experiments. Nitrous oxide flux reached a peak in the sand cores at 6–9 days after nitrate was applied (Table 5). The N_2O flux peak at 20 and 30 cm occurred 3–5 days after the pulse of peak nitrate concentration passed the 20 and 30 cm depths (Table 5). The first N_2O flux peak was followed by a steep decline in N_2O flux. Flux then increased to a peak followed by a decline. The second and subsequent N_2O flux peaks were never as high as the initial N_2O flux peaks for two reasons. First, N_2O reduction was no longer inhibited by acetylene after 7–10 days so that some of the N_2O may well have been reduced to N_2 . Second, after 10 days, about 1.5 volumes of soil moisture had been leached through the sand cores, leaving little residual nitrate. Denitrification activity declined as the NO_3^- concentration in the soil water decreased.

Table 6 shows the cumulative N lost by denitrification at the time of peak N_2O flux for each depth in the sand cores. Only 1–2% of applied N was lost by denitrification after 7–10 days for experiments 2–5. Significant N loss occurred only in core 1, experiment 6 where 9.5% of applied N was lost to denitrification in the first 10 days. The cumulative N loss to denitrification at the end of each sand experiment is listed in Table 7.

The values in Table 7 are minimum values for total N loss because any N_2 flux occurring after 10 days was not measured. One of the greatest uncertainties in recent research on denitrification in soils is the relative proportion of N_2O and N_2 produced. The availability of organic C or other electron donor compounds and the rate of denitrification are among the factors that influence the ratio of N_2O to N_2 (Firestone, 1982; Egginton and Smith, 1986b). Increasing C availability generally increases the proportion of N_2 produced (Smith and Tiedje, 1979). Similarly, the ratio of N_2O to N_2 has been found to decrease as the rate of denitrification increases (Egginton and Smith, 1986b) or when very wet soil conditions inhibit the diffusion and release of N_2O (Rolston and Marino 1976; Rolston et al., 1980). The ratios of N_2O to N_2

Table 7
Minimum N loss by denitrification during each experiment

Experiment	Core number	Nitrogen loss (%)
Sand 2	1	2
Sand 2	2	1
Sand 3	1	2
Sand 3	2	1
Sand 4	1	2.5
Sand 4	2	1
Sand 5	1	3
Sand 5	2	2
Sand 6	1	17
Sand 6	2	4

emissions reported in the literature are extremely varied. For example, Mosier et al. (1986) found that 30–50% of gaseous N emissions from field microplots were N_2 . Egginton and Smith (1986a,b) report ratios of N_2O/N_2 ranging from 0.2 to 1 and Dowdell and Webster (1984) report 100% N_2 emissions.

Total N_2O flux calculated for the various depths in the sand cores did not correspond with volumetric water content and depth as expected. Fig. 14, which shows total N_2O flux for sand experiment 3, core 1 is a representative example. Although the 20 cm and 30 cm depths had average volumetric water contents equal to 20% and 27%, respectively, the N_2O flux at these depths is much lower than the flux at 10 cm ($\theta \approx 8\%$) and also at the surface at most times. The N_2O flux at 30 cm exceeded the flux at 10 cm only during peak flux at 216 h however, at this time, the acetylene blockage was no longer effective. All of the N_2O flux plots (data not shown) exhibit to some degree the same lack of correspondence between N_2O flux and water content. Two factors contribute to this effect. First, during time 0–196 h, the N_2O flux at the surface and at 10 cm exceeded the flux at 20 and 30 cm partly because the peak nitrate pulse had not yet reached the lower depths. As Table 3(a) shows, there is a lag time of 3–6 days (72–144 h) from the time the nitrate pulse passed the 10 cm depth until it reached the 30 cm depth. The second factor is the result of a significant partitioning of water content with depth that occurs in each sand experiment. Since N_2O is quite soluble in water, much of the N_2O evolved at 20 and 30 cm was dissolved in the soil solution, diffused through the liquid-filled pore space, and finally diffused into the soil air space in the drier, upper portion of the cores. The net effect was that, in experiment 3, the flux measured at 10 cm or at the surface, was often greater than that at 30 cm. Only during experiment 6 did the N_2O flux at 30 cm exceed the measured flux at the shallower depths (Fig. 15). In experiment 6, denitrification activity appears to have been enhanced by the presence of an established root zone. Although the water content was highest at the 30 cm depth in both sand cores, more N_2O was generated than could dissolve into the soil solution, so the measured flux at 30 cm exceeded the flux at the shallower depths. Also, the flux at 20 cm was negative which indicates a flux upwards into the overlying layer. The clay cores showed the same relationship subsequently; this will be discussed later in the denitrification section for clay cores.

3.5. Denitrification: sandy soil

Very little nitrogen was lost as the gaseous end products of denitrification from the unplanted sand cores. This result was not unexpected since the sandy soil was coarse textured, well drained, and had a low organic matter content. Because a fallow sandy soil has little tendency to denitrify under unsaturated conditions, almost all the nitrate applied to the tilled infiltration basins at the Reedy Creek Improvement District will be leached rapidly. While the nitrogen loss owing to denitrification in the unplanted sand cores is small, only about 1% during the first 7–10 days after application, the amount is not negligible. Over the duration of the various unplanted sand experiments, the minimum nitrogen lost as N_2O was 2%. Since acetylene inhibits the conversion of N_2O to N_2 , then N_2O fluxes are indicative of total denitrification

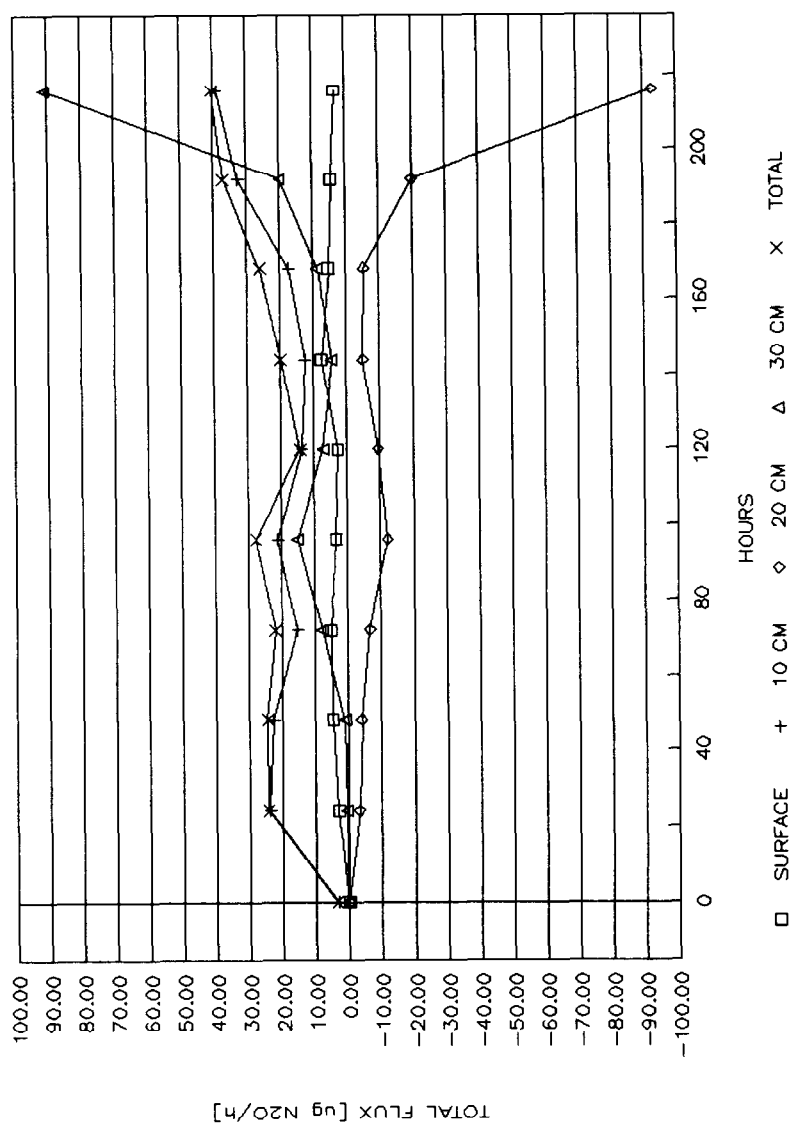


Fig. 14. N₂O flux: core 1, sand experiment 3.

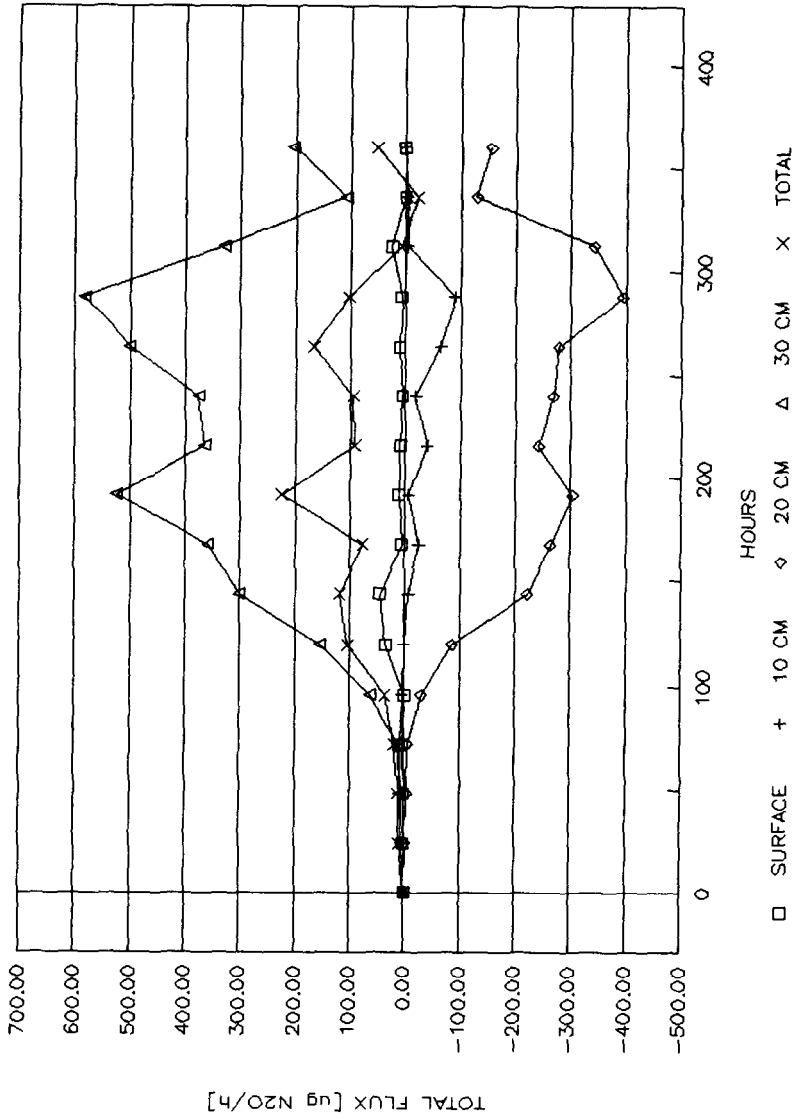


Fig. 15. N_2O flux; core 1, sand experiment 6.

losses. Nevertheless, the sandy soil can be expected to be subject to bursts of denitrification activity after a rainfall or irrigation event. The most surprising result was the sharp increase in N_2O emissions that occurred in the planted experiments. The tomatoes were planted at the start of experiment 5 and only a slight increase in denitrification was observed. However, by experiment 6, the plants were established for over a month and denitrification activity increased significantly. Core 1 lost almost 10% of applied N to denitrification in the first 10 days. Core 2, with the greater permeability and generally lower water content, lost only 2% in 10 days. The total N lost as N_2O after three weeks was 17% for core 1 and 4% for core 2. This is in contrast to the findings of Haider et al. (1987); actively growing plants had no effect on denitrification rates and denitrification did not increase until plant senescence. The results of experiment 6 suggest that the sandy soil might experience significant denitrification under unsaturated conditions when planted in vegetable crops because vegetable crops are shallow rooted, generally irrigated, and heavily fertilized. Although the presence of tomato plants in both sand cores stimulated denitrification activity because carbon derived from exudates or sloughed off root cells provided the carbon necessary for a high rate of denitrification, the net effect appears to be very sensitive to soil moisture content. So, while the planted sand soil has a surprisingly great potential for denitrification under unsaturated conditions, net denitrification activity in planted fields may be controlled by small spatial and temporal variations in soil moisture content.

The ratio of N_2O to N_2 emitted by the sandy soil during denitrification was not determined. Assuming that 20–50% of the gaseous N emissions occur as N_2O (Egginton and Smith, 1986a; Mosier et al., 1986), the sandy soil could lose from 0.5 to 10% of applied fertilizer N as N_2O emissions during the first few weeks after application. Eichner (1990) estimated that about 3% of the fertilizer N applied worldwide is emitted as N_2O . Previous studies of denitrification in agricultural soils under unsaturated conditions confirm that significant gaseous nitrogen emissions can occur within a few weeks of fertilizer application. Kowalenko (1980) found that 28% of fertilizer N applied during early summer to a fallow sandy soil microplot was lost to denitrification within 35 days. Subsequent gaseous N loss was insignificant. Dowdell and Webster (1984) and Webster and Dowdell (1985) reported that total gaseous N losses during two long-term studies ranged from 2 to 19% of applied nitrogen which is close to that lost during this study.

3.6. N_2O flux: clay cores

Figs. 16 and 17 show total N_2O flux for clay cores A and B. Total N_2O flux and cumulative N_2O flux (data not shown) for the clay cores were vastly lower than the N_2O flux from the sand cores. In core A, only 1.5 mg N or 0.1% of applied N was lost as N_2O 240 h after fertilization. In core B, only 0.39 mg N or 0.03% was lost after 240 h. After 984 h or 41 days, N_2O emissions from core A accounted for only 0.6% of the fertilizer N applied. After 792 h or 33 days, the gaseous N loss from core B as N_2O was only 0.1%.

The largest N_2O flux in core A occurred at 30 cm followed by the surface, 20, and

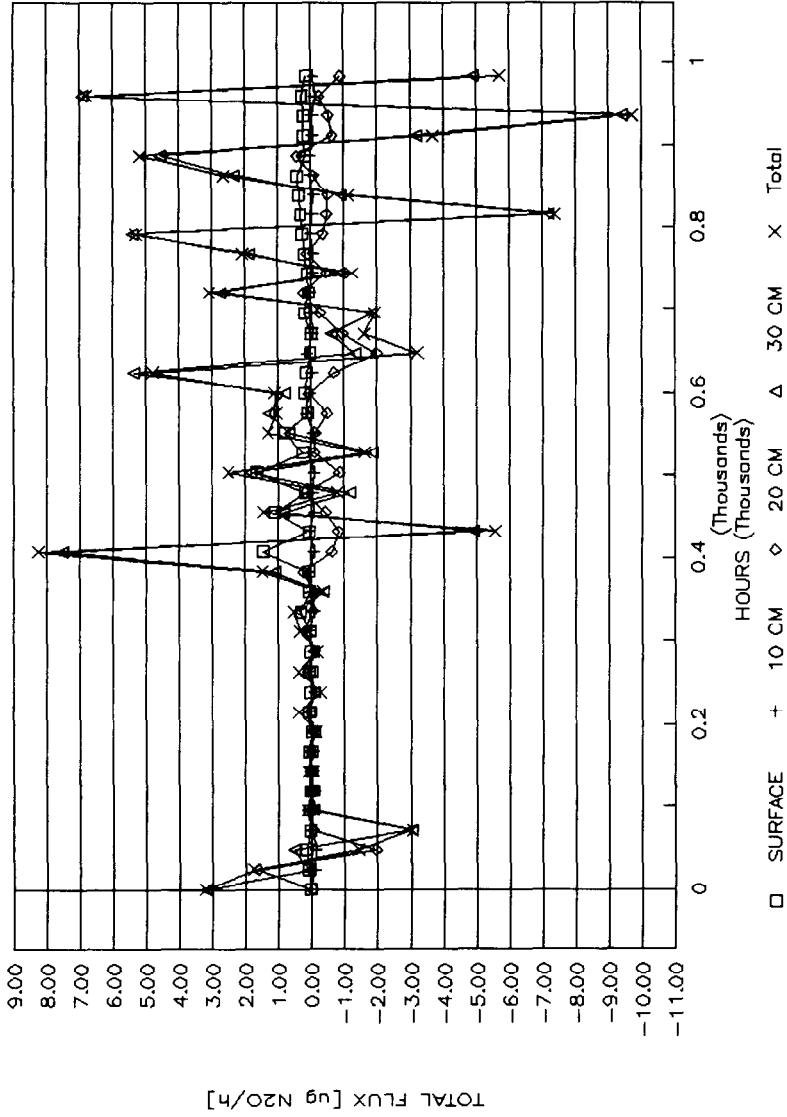


Fig. 16. N₂O flux; clay core A.

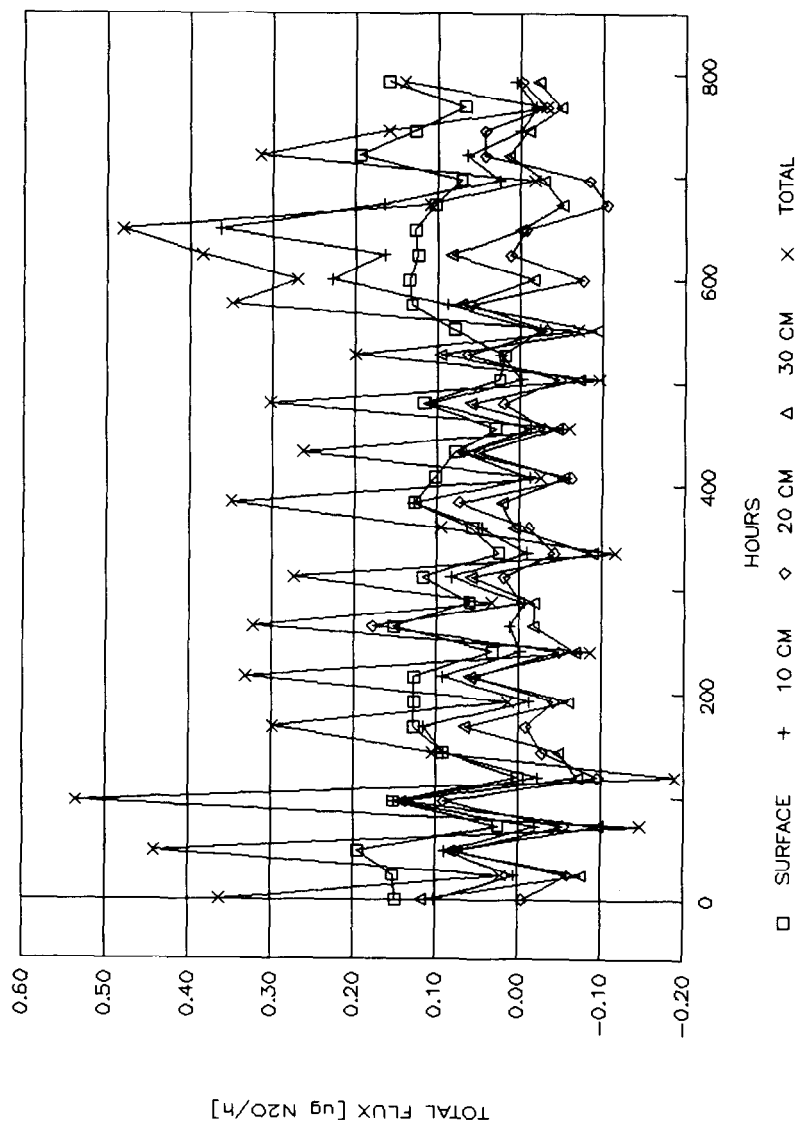


Fig. 17. N₂O flux; clay core B.

10 cm. In core B, the largest flux occurred at the surface, the least flux at 20 cm. Acetylene was applied to the clay cores three days before the calcium nitrate was applied to allow time for the acetylene to diffuse through the cores. No acetylene was found to be present in any of the gas samples analyzed from core B; thus, it appears that no acetylene inhibition occurred in core B. The acetylene was most likely metabolized by microorganisms, particularly after several days of exposure. In any event, the low N_2O flux from core B (Fig. 17) primarily is due to the lack of acetylene inhibition and is likely not indicative of flux from the clay soil.

Fig. 16, core A, shows that a very early N_2O flux peak occurred from 0 to 48 h at 30 cm, apparently the result of immediate channeling of nitrate to the 30 cm depth via preferential flow. By 72 h, flux at 30 cm and other depths in core A was extremely low, less than $0.50 \mu\text{g h}^{-1}$, suggesting little denitrification activity. Then, at 408 h or 17 days, N_2O flux resumed and fluctuated widely at 30 cm; this continued until the end of the experiment at 984 h. While the increase in the surface N_2O flux in core A coincided with the increase in N_2O flux at 30 cm, the surface N_2O flux peaks were offset from the peaks at 30 cm by 24–48 h. The surface flux of N_2O was caused by diffusion of N_2O from 30 cm up to the surface along preferential flow paths, but was minimal in comparison to the sand cores (Figs. 14 and 15). There was virtually no N_2O flux at 10 cm and the flux at 20 cm was negative which indicates diffusion of N_2O upwards from the 20 cm layer to the 10 cm layer. Apparently, denitrification activity at depths 0–30 cm contributed little to surface N_2O flux in core A. When the volumetric water content for core A at 30 cm ($\theta = 0.36$ to 0.42) is compared with Fig. 16, an interesting relationship between volumetric water content and N_2O flux is evident. As discussed in the previous section on nitrate leaching, soil water seemed to accumulate in a tight zone in core A at about 30 cm. Once the volumetric water content at 30 cm exceeded 40%, soil water was able to infiltrate the tight zone and volumetric water content at 30 cm decreased during the next 24–72 h. Consequently, denitrification activity at 30 cm was stimulated when the water content exceeded 40%; after 24–48 h, an N_2O flux peak developed. The cyclic buildup of water content at 30 cm followed by an N_2O flux peak 24–48 h later, continued until the end of the experiment in core A and suggests that no significant denitrification activity occurred at volumetric water contents less than 41% in this soil.

3.7. Denitrification: clay soil

The clay soil was expected to be more susceptible to denitrification than the sand soil because of the greater moisture content in the clay, the longer residence time of nitrate in the clay, and restricted oxygen diffusion in the soil beyond the soil macropores. Instead, the nitrous oxide emissions from the clay cores were one to two orders of magnitude less than the emissions from the sand cores. The low N_2O emissions may result, in part, from incomplete acetylene inhibition in the clay cores owing to rapid leakage of acetylene through macropores in the soil. The acetylene inhibition technique does not appear to be effective on structured clay soils. Because no acetylene was measured in the soil gas samples obtained from core B, it was concluded that no acetylene inhibition was achieved in core B. Acetylene inhibition was achieved

for the first 5–7 days in core A; after that time, no acetylene was present in the soil gas samples. During the first 10 days after nitrate was applied, only 0.1% of applied N was lost from core A as N_2O and the N_2O emissions from core B were negligible. Total N emitted as N_2O from core A was 0.6% after 6 weeks. Total N emitted as N_2O from core B was 0.1% after nearly 5 weeks. Despite possibly limited acetylene inhibition, the extremely low N_2O emissions indicate that much less denitrification occurred in the clay cores than in the sand cores. Even if nitrogen loss as N_2 was as much as four or five times the loss as N_2O , the apparent denitrification activity in the clay soil was still significantly less than the denitrification activity in the sand soil. Mosier et al. (1986) found that the volatile N loss from a clay loam soil (32% sand, 32% silt, 36% clay) was 2.5% of N applied to a corn field and about 1% of N applied to barley. In the former case, 70% of emissions occurred as N_2O and 30% as N_2 ; in the latter case the N_2O and N_2 emissions were approximately equal. Large gas flux occurred only after water additions by either irrigation or rainfall. Mansell et al. (1986) determined that the gaseous N loss was negligible from an irrigated Florida sand soil planted in citrus.

The extremely low N_2O flux observed from the clay cores during this experiment is similar to the results obtained by Craswell and Martin (1975a,b) from a nitrogen balance study on a cracking clay soil with low organic carbon content. Although Craswell and Martin (1975a,b) did not measure soil emissions directly, 96–98% of ^{15}N was recovered from the soil, plant tissue, and leachate after 1.5 weeks. No significant gaseous nitrogen loss was observed unless the clay soil was saturated. Results of this study indicate that well-structured, cracking clay soil is not especially prone to denitrification unless saturated and subject to high soil temperatures such as might occur following a summer storm.

Recent work by Booltink and Bouma (1991) provides some insight into why the clay soil in this experiment may be less susceptible to denitrification activity than well-drained, coarser textured soils. They observed flow patterns and processes in well-structured clay soil cores and demonstrated that almost all infiltration occurred after internal catchment of water in the bottom of discontinuous macropores. In the cores they studied, about 40% of the soil volume was affected by infiltration after internal catchment. Another one third of the soil volume was in contact with continuous macropores yet received little lateral infiltration of water moving through these macropores under unsaturated conditions. Only about one third of the water applied to the cores infiltrated the soil peds after internal catchment. More than half the applied water moved completely through the cores along the continuous macropores; only a few percent of applied water infiltrated the soil laterally from continuous macropores which would inhibit denitrification within a majority of the soil volume. Greenwood (1975) theorized that a soil must develop a continuous water film at least 0.9 mm thick over the soil particle surfaces before denitrification can occur. Booltink and Bouma (1991) demonstrated that continuous water films are unlikely to develop along preferential flow paths in a structured clay soil except under saturated conditions when the macropores are filled with water. Water saturated zones in an otherwise well-drained soil are more likely to develop in an unstructured soil such as the sandy soil in this study as a result of uneven water distribution.

The concept of slow and discontinuous infiltration after internal catchment may explain some of the results obtained for the clay cores during this project. Nitrate was leached more slowly and less total nitrate was leached from the clay cores because some portion of the applied nitrate was trapped at the bottom of discontinuous macropores. Similarly, little apparent denitrification activity occurred in the clay because denitrification was probably restricted to the surfaces of the cores immediately following water application plus a relatively small volume of soil receiving infiltration from internal catchment areas. Less denitrification occurred in core B than core A because core B has more extensive preferential flow and correspondingly less internal catchment than core A. The susceptibility of a clay soil to denitrification under unsaturated conditions may depend upon the relative amount of preferential flow through continuous macropores. Thus, structured clay soils might be characterized by a continuum of susceptibility to denitrification determined by the morphological characteristics of the soil pore system.

Figs. 16 and 17 show a very low flux (negative) for the 20 cm depths in the clay soil when compared with the 30 cm depth (positive) (this is also true for the sand cores in Figs. 14 and 15). When the flux is negative, as at the 20 cm depth, it signifies upward movement into the overlying layer and when positive, as at the 30 cm depth, downward movement is indicated. When comparing this to two different models (Millington, 1959; Currie, 1970) at various effective porosities (Fig. 18), it appears that at low water content (Table 2), there is good correlation between the soil

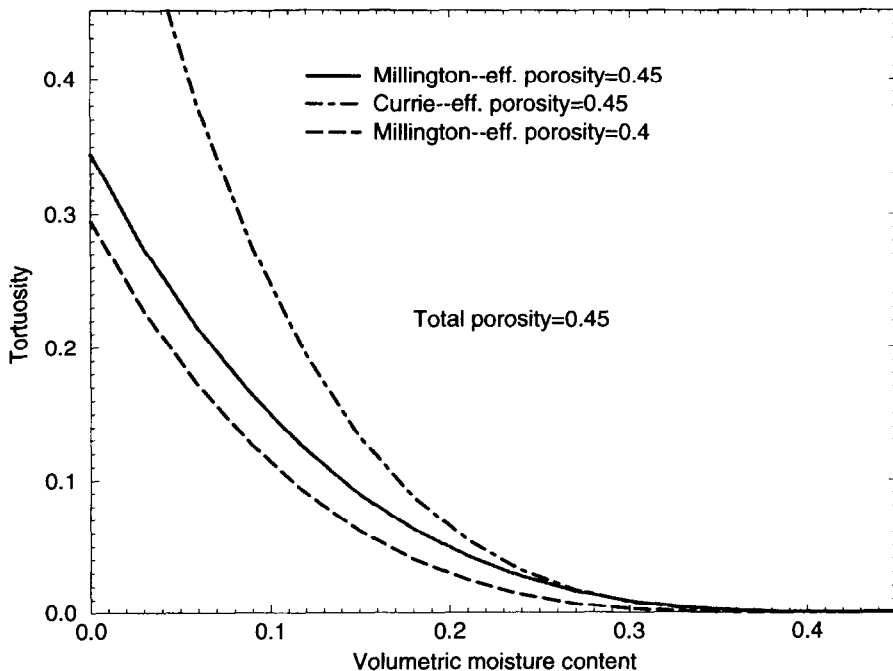


Fig. 18. Comparison of tortuosity (ξ) with volumetric water content by three models.

tortuosity (ξ) and water content and thus, flux. However, at high water content (Table 2), tortuosity is overestimated; this would cause a small overestimation in flux based on equation 7. As a result, the data of Figs. 14–17 show basically offsetting curves for the 20 and 30 cm depths for both soils. For the sand cores, in combination with the overestimation of tortuosity at high water content, denitrification conditions may occur at the 30 cm depths while transformation processes are shifted towards nitrification at the 20 cm depths, as indicated by the volumetric water contents in Table 2. For the clay cores, it is likely that there was not enough surface area contact of soil to solution for great amounts of denitrification to occur at the 20 cm depth versus the 30 cm depth; this is evidenced by the stained areas of the cores (Figs. 9 and 10) and the calculated fluxes (Figs. 16 and 17). This is especially true for core B which had a significantly higher water content at 30 cm than at 20 cm (Table 2) and is in agreement with the discussion in the preceding paragraph.

4. Conclusions

Results from both the sand and clay experiments show that an understanding of the dynamics of nitrate transport coupled with denitrification must include the following factors: small variations in the soil moisture content profile, gaseous diffusion coefficients for the soil as a function of soil moisture content, the rate at which the nitrate pulse is moving through the soil, the solubility of N_2O and N_2 , and diffusion coefficients of these gasses through the soil solution. For example, denitrification activity in the clay cores appears to be very sensitive to small variations in soil moisture content. In the sand cores, the correspondence of N_2O emissions with depth and soil water content is confounded by the solubility of N_2O in soil pore water at depth. It appears that much of the N_2O diffused through the soil solution and came out of solution as N_2O gas in the upper parts of the cores where the volumetric water content was lower. Previous research on nitrate transport and transformation in the unsaturated zone has not included the combined effect of N_2O solubility in water and the development of a water content profile in the soil.

Perhaps the most unexpected result was that very little denitrification activity, as indicated by N_2O emissions (one to two orders of magnitude less than the sand cores without plants), was observed in the clay soil despite the greater moisture content of the clay and the retarded leaching of nitrate in the clay. Under unsaturated conditions, this clay soil seems to have little or no tendency to denitrification. In the light of recent research into flow processes in structured soils exhibiting preferential flow, limited denitrification in the clay soil is attributed to the limited volume of soil available for infiltration of pore fluids and the development of denitrifying conditions. Perhaps solubility of N_2O in the soil solution and very slow diffusion of N_2O through the low permeability soil pore network are also factors in the low N_2O emissions from the clay cores. Nitrous oxide may also be retained within the soil solution for a considerable time so that the emissions from the clay soil must be observed over a far longer period than was done in this experiment to quantify N loss owing to denitrification activity accurately. A more thorough

experiment using a labelled fertilizer technique could be conducted on the clay cores the better to quantify denitrification activity in the clay and to relate denitrification activity to water content and to the characteristics of the preferential flow network.

The sandy soil exhibited a surprising amount of rapid denitrification when planted with tomatoes with fluxes several orders of magnitude greater than for the clay cores and also for sand cores without plants. This result should be examined further since few, if any, studies of denitrification and nitrogen balance have been conducted on soils planted with vegetable crops. Vast tracts of sandy soil are planted all the year round with irrigated vegetable crops in central and southern Florida. This project suggests that the amount of fertilizer N emitted as N_2O could range from as little as 0.5% of the N applied to over 10%; in the latter case the sandy soil could be a significant source of N_2O emissions.

Leaching loss of surface-applied NO_3^- -N was significant in both the sandy and the clay soils. In the sand cores, most of the applied nitrate was leached below the rooting zone within 10 days. Leaching occurred less rapidly in the clay cores; nevertheless, about 60% of the nitrate-N was leached from the unplanted clay soil in 5–6 weeks. This is probably because sandy soils have greater drainage volumes owing to the low water-retaining capacity. Under low intensity watering, water and nitrate move into the soil aggregates or peds in the structured clay soil and less leaching of nitrate occurs. Unstructured sandy soils have no soil aggregates to retain nitrate and, consequently, leaching losses are much greater. The experiment should be repeated with planted clay cores to determine how plant uptake influences leaching from the clay. Based on the results of this experiment, the waste water infiltration basins at Reedy Creek Improvement District present a potential aquifer contamination hazard depending upon the concentration of nitrate in the water applied to the basin. Likewise, agricultural practices on this soil in the surrounding area, which use heavy fertilization and irrigation also pose a threat to groundwater quality.

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